Master of Data Analytics

Improving Public Service Resolution Efficiency: An Analytics Study of NYC 311 Service Requests (2025–2026)

Project Report

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A proposal submitted to

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In Fulfilment of the Requirements for the Course

DAMO-699-7: Capstone Project

Winter 2026

Signature Page

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ABSTRACT

Efficient public service delivery is essential for maintaining urban infrastructure and improving citizen satisfaction. Municipal governments receive a large number of non-emergency service requests, making it challenging to monitor operational performance and ensure timely resolution. This study analyzes the NYC 311 Service Request dataset to identify factors influencing service resolution efficiency and to examine patterns associated with delayed service resolution.

The dataset used in this study contains approximately 149,000 service request records collected from the NYC Open Data portal for the period January 2025 to January 2026. Data preprocessing and cleaning were conducted using Python to ensure data quality and to construct analytical variables such as resolution time, delay classification, temporal indicators, and agency workload metrics.

Several analytical methods were applied to investigate the research questions. Statistical techniques including the Kruskal–Wallis test and Chi-Square tests were used to examine differences in resolution time and associations between categorical variables and delay probability. Predictive modeling techniques such as logistic regression, decision tree classification, and random forest models were implemented to evaluate whether delayed service resolution can be predicted using historical service request attributes. In addition, Power BI dashboards were developed to visualize service demand, geographic patterns, operational trends, and delay distributions.

The results show that resolution performance varies significantly across boroughs, agencies, complaint types, and time periods. Infrastructure-related complaints, particularly street lighting issues, exhibit the highest delay probability. Predictive models demonstrate strong capability in identifying delayed service requests, with model performance achieving ROC-AUC

values above 0.85. The findings also indicate that higher agency workload significantly increases the likelihood of service delays.

Overall, the study demonstrates how data analytics and predictive modeling can support more efficient municipal service management and improve operational decision-making in urban service systems.

ACKNOWLEDGMENTS

We would like to express our sincere gratitude to our faculty advisor, Professor Mohamad Fawaz, for his guidance, valuable feedback, and continuous support throughout the completion of this capstone project. His insights and encouragement greatly contributed to the development and success of this research.

We also thank the University of Niagara Falls Canada and the Master of Data Analytics program for providing the opportunity and academic resources necessary to conduct this study.

In addition, we acknowledge the NYC Open Data portal for providing access to the NYC 311 Service Request dataset, which served as the foundation for this research and enabled us to conduct meaningful analytics on public service performance.

Finally, we would like to thank all the members of Group 5 for their collaboration, dedication, and teamwork in completing this project successfully.

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1. CHAPTER 1: INTRODUCTION

1.1 PROBLEM IDENTIFICATION

Efficient public service delivery is a key responsibility of municipal governments. In large cities, managing services such as water supply, street lighting, sanitation, and infrastructure maintenance is challenging due to high population density and large volumes of citizen requests.

The NYC 311 Service Request System allows residents to report non-emergency issues, which are then routed to the appropriate city agencies for resolution. These requests generate large amounts of operational data that can be used to evaluate public service performance.

This project analyses 311 service request data from January 2025 to January 2026 to identify patterns in service delays across boroughs, agencies, and complaint types. Using descriptive analytics, statistical testing, and predictive modeling, the study aims to provide insights that support more proactive and data-driven municipal service management.

1.2 RESEARCH QUESTIONS AND HYPOTHESIS

The primary objective of this study is to evaluate service resolution efficiency within the NYC 311 system and identify factors associated with delayed service resolution. In addition, the study examines whether predictive analytics techniques can be used to identify service requests that are likely to experience delays.

Based on this objective, the following research questions guide the analysis:

1. How do service request volumes and resolution times vary across boroughs, agencies, and complaint types?
2. Which agencies and complaint types exhibit higher probabilities of delayed service resolution?

3. What temporal factors, such as weekday patterns and seasonal variation, influence resolution performance?
4. Can delayed service resolution be predicted using historical service request attributes?
5. How does agency workload influence the likelihood of delayed service resolution?

To address these research questions, the following hypotheses were formulated:

- **H_{01}:** There is no significant difference in resolution time across boroughs.
- **H_{11}:** Resolution time differs significantly across boroughs
- **H_{02}:** Complaint types and agencies do not significantly affect the probability of delayed resolution.
- **H_{12}:** Certain complaint types and agencies exhibit significantly higher delay probabilities.
- **H_{03}:** Temporal factors such as weekday/weekend and seasonal variation do not significantly influence delayed resolution probability.
- **H_{13}:** Temporal factors significantly influence delayed resolution probability.
- **H_{04}:** Historical request attributes do not significantly predict delayed service resolution.
- **H_{14}:** Historical request attributes significantly predict delayed service resolution.
- **H_{05}:** Agency workload has no significant effect on the likelihood of delayed resolution.
- **H_{15}:** Higher agency workload significantly increases the likelihood of delayed service resolution.

These hypotheses are tested using statistical analysis and predictive modeling techniques to determine the operational drivers of delayed service resolution in the NYC 311 system.

2. CHAPTER 2: DATA CLEANING AND PREPARATION

2.1 DATA COLLECTION

The dataset used in this study was obtained from the NYC Open Data portal, which provides publicly accessible records of non-emergency municipal service requests submitted through the NYC 311 Service Request System. The working dataset, 311_Service_Requests.csv, contains approximately 149,000 service request records from January 2025 to January 2026. Each record represents a complaint submitted by residents and includes operational, temporal, and geographic attributes describing the service request lifecycle. Key variables include the Unique Key (request identifier), Created Date, Closed Date, Agency, Borough, Problem (formerly Complaint Type), Problem Detail (Descriptor), Status, Resolution Description, and geographic location information such as address, latitude, and longitude. These variables enable analysis of service resolution time, complaint patterns, borough demand distribution, agency workload, and temporal trends.

After loading the dataset, a working copy was created from the raw file to preserve the original data and ensure that cleaning and transformation were performed on a separate analytical version.

The dataset includes service-request level fields such as unique request identifier, created date, closed date, agency, borough, complaint type, problem detail/descriptor, address and location information, resolution description, and status. These variables support exploratory data analysis, hypothesis testing, delay modeling, and dashboard development. A structured summary of the dataset attributes is provided in (*See* ***Table 1****:* Description of NYC 311 Service Request Dataset Attributes).

In addition, a separate **metadata file** was created to document the dataset structure, variable definitions, and data types. This metadata file provides a detailed description of each attribute and supports transparency, reproducibility, and ease of interpretation during analysis.

2.2 DATA CLEANING

Data cleaning was performed systematically in Python to improve consistency, reduce noise, and prepare the dataset for both statistical analysis and Power BI visualisation. The cleaning process followed a structured sequence.

2.2.1 CREATING A WORKING COPY OF THE DATASET

The raw CSV file was first loaded into a dataframe, and a deep copy of the data was created for preprocessing. This step protected the original dataset from accidental modification and improved the reproducibility of the data pipeline.

2.2.2 REMOVING UNNECESSARY COLUMNS

Several columns were removed because they were either highly sparse, operationally irrelevant to the hypotheses, duplicated other location information, or not useful for modeling and dashboard reporting. These included fields such as agency name, cross streets, landmark, facility type, due date, resolution action updated date, bridge-related attributes, vehicle type, and state-plane coordinate variables. This reduction step improved analytical focus by retaining variables directly relevant to complaint resolution, location, timing, and agency performance.

2.2.3 MISSING VALUE ANALYSIS

A missing-value summary was generated to inspect null values across the dataset. During this step, **Closed Date** was intentionally excluded from the initial null summary because missing

values in that field were expected for open or unresolved requests rather than representing data errors.

This distinction was important because the project examined both:

- **closed requests**, for resolution-time analysis, and
- **open requests**, for backlog and elapsed-time analysis.

2.2.4 HANDLING MISSING VALUES

Missing values were handled using domain-based rules rather than blanket deletion.

The following text and address-related columns were filled with "Unknown" where missing: Street Name, Incident Address, City, and Location.

In addition, the **Resolution Description** field was filled with **"Not Provided"** when missing.

The **Incident Zip** field was explicitly converted to string format, stripped of extra spaces, and then filled with **"Unknown"** where required. This was important because postal code values behave more appropriately as identifiers than as numeric variables.

2.2.5 CHECKING FOR DUPLICATE RECORDS

The dataset was checked for duplicate records using the **Unique Key** field, which served as the primary identifier for each service request. Duplicate verification was necessary to ensure that each request was counted only once in volume summaries, closure analysis, and predictive modelling.

2.2.6 CONVERTING DATE FIELDS TO DATETIME FORMAT

Created Date and Closed Date were converted to datetime format using pd.to_datetime() to enable time-based analysis and safely handle invalid dates as missing values.

2.2.7 CONVERTING CATEGORICAL VARIABLES TO STRING TYPE

Operational and geographic variables (e.g., Agency, Problem, Incident Address, City, Status, Borough, Location) were cast to string type to standardise categories for analysis and visualisation.

2.2.8 CLEANING TEXT VALUES

Leading and trailing whitespace was removed from categorical fields to prevent duplicates and ensure consistent category counts for analysis and visualization.

2.2.9 CALCULATING RESOLUTION TIME

Created resolution_hours (Closed Date − Created Date) and resolution_days for interpretability, supporting delay classification, outlier detection, borough and complaint-type analysis, and hypothesis testing.

2.2.10 REMOVING INVALID RESOLUTIONS

Removed records with negative resolution_hours, keeping only missing or non-negative values to exclude logically invalid closed requests while retaining open ones.

2.2.11 CREATING CLOSURE STATUS

Created binary is_closed (1 = closed, 0 = open) to distinguish resolved versus unresolved requests for EDA.

2.2.12 CALCULATING ELAPSED TIME FOR OPEN REQUESTS

Computed elapsed_hours_so_far for open requests using a fixed reference date (AS_OF = 2026-02-10) to analyse backlog and pending durations.

2.3 EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to examine the distribution, volume, and operational patterns of NYC 311 service requests. The analysis focused on resolution time, open request duration, and complaint volume.

2.3.1 RESOLUTION TIME ANALYSIS (CLOSED CASES)

The resolution time (hours) for closed service requests was analyzed using a boxplot and distribution plots. The boxplot shows a wide spread with several extreme values, indicating the presence of long-running cases. These outliers were retained as they represent real operational scenarios.

The histogram confirms that resolution time is highly right-skewed, with most requests resolved quickly and a smaller number taking significantly longer. A log-scaled distribution was also used to improve interpretability by compressing extreme values and revealing the overall pattern more clearly *(See **Figure 1:** Boxplot of Resolution Time (Hours) — Closed Cases Only) (See **Figure 2:** Distribution of Resolution Time (Hours) — Closed Cases Only), (See **Figure 3:** Log-Scaled Distribution of Resolution Time (Hours) — Closed Cases Only)*

2.3.2 OPEN SERVICE REQUEST ANALYSIS

The duration of unresolved cases was examined using elapsed time for open service requests. The distribution shows that many open cases persist for extended periods, indicating backlog accumulation.

A log-transformed version of elapsed time was used to better visualize long-duration cases and understand variation across open requests. Additionally, the median backlog duration was calculated to summarise the typical waiting time for unresolved requests *(See **Figure 4:** Elapsed*

*Time for Open Service Requests), (See **Figure 5:** Log Elapsed Time for Open Service Requests), (See **Figure 6:** Median Backlog Duration).*

2.3.3 COMPLAINT TYPE ANALYSIS

Service request volume was analysed by complaint type to identify the most common issues reported by citizens. The results show that certain complaint categories dominate the dataset, indicating higher service demand in specific areas *(See **Figure 7:** Complaint Types by Volume), (see **Figure 8:** Service Request by Borough)*

2.4 FEATURE ENGINEERING

Feature engineering transformed raw administrative variables into analytically meaningful features that supported statistical analysis and predictive modeling.

- **Resolution Time Features**

  Two key variables were created to measure service resolution performance:

  1. **resolution_hours** – resolution time measured in hours
  2. **resolution_days** – resolution time measured in days

  These features formed the foundation for delay analysis and hypothesis testing.

- **Open Case Aging Feature**

  - For unresolved service requests, the feature **elapsed_hours_so_far** was created to measure how long the request had been open as of a defined analysis date.
  - This variable enabled the study to evaluate the unresolved service backlog.

- **Delay Classification Feature**

  - A binary delay variable called **is_delayed** was created to classify service requests based on resolution duration.

- The delay threshold was determined using the **80th percentile of resolution time**, meaning that the top 20% longest service requests were classified as delayed.
- This threshold was selected after sensitivity testing using the 75th and 90th percentiles to ensure a balanced classification distribution.

❖ **Temporal Features**

- Temporal features were generated to capture time-based service patterns. These include, request creation day, weekday indicator, weekend indicator and seasonal categories
- These features enabled the study to evaluate temporal influences on service delays.

❖ **Agency Workload Feature**

- A key operational feature called **agency_workload_daily** was created to measure the number of active service requests assigned to each agency on a given day.
- This feature was constructed by calculating the cumulative number of open requests per agency using daily event aggregation.
- The workload variable served as an operational proxy for agency demand and was later used to test the hypothesis that higher agency workload increases the probability of delayed service resolution.

2.5 RELATIONAL DATABASE DESIGN AND POWER BI INTEGRATION

A relational database was developed using SQLite to efficiently organize the cleaned NYC 311 service request dataset. The database was designed using a star schema structure, consisting of a central fact table (fact_requests) connected to multiple dimension tables (dim_agency, dim_borough, dim_problem_type, and dim_time). The fact table stores key performance measures such as resolution time, delay indicators, and agency workload, while the dimension tables provide

descriptive attributes that enable multi-dimensional analysis. This design improves data consistency, reduces redundancy, and enhances query performance, making it suitable for analytical and reporting purposes.

The relational database was then integrated with Power BI to support interactive data visualisation and business intelligence reporting. The structured tables and SQL views were connected directly to Power BI, enabling efficient data retrieval and transformation. Additionally, aggregated datasets were exported as CSV files to optimize performance for complex visualisations such as maps, time-series trends, and comparative analysis. This integration allowed the development of dynamic dashboards that provide insights into service request patterns, resolution efficiency, and delay behaviour across agencies, boroughs, and time periods.

3 CHAPTER 3: STATISTICAL ANALYSIS AND DIAGNOSTICS

This study applies a combination of statistical and machine learning techniques to analyse service resolution performance and predict delays in NYC 311 service requests. The selected methods were chosen based on the nature of the data, research objectives, and the need to capture both statistical relationships and predictive patterns.

The following analytical methods were used:

1. **Kruskal–Wallis Test**

   A non-parametric statistical test is used to compare resolution time distributions across boroughs, as the data is non-normally distributed.

2. **Chi-Square Test of Independence**

   Applied to examine associations between categorical variables such as agency, complaint type, and delay status.

3. **Logistic Regression**

   Used as a baseline predictive model to estimate the probability of delayed service resolution and interpret the influence of key variables.

4. **Decision Tree Classification**

   Implemented to capture nonlinear relationships and provide interpretable feature importance for delay prediction.

5. **Random Forest Model**

   Applied to improve predictive performance and capture complex interactions between variables.

These methods collectively provide both statistical validation and predictive insights, enabling a comprehensive analysis of service performance and delay drivers.

3.1 HYPOTHESIS 1: DIFFERENCES IN RESOLUTION TIME ACROSS BOROUGHS

3.1.1 METHODS APPLIED

This hypothesis examined whether service request resolution times differ across the five boroughs of New York City: Bronx, Brooklyn, Manhattan, Queens, and Staten Island. Resolution time was calculated as the difference between the Created Date and Closed Date, stored as the variable resolution_hours.

The dataset was filtered to include only closed service requests with valid resolution times and non-missing borough values, while records labelled “Unspecified” were removed. Because resolution time data is typically right-skewed and may contain extreme values, a non-parametric statistical approach was applied.

The Kruskal–Wallis H test was used to determine whether resolution time distributions differ significantly across boroughs. When significant differences were detected, a Dunn post-hoc test with Bonferroni correction was conducted to identify which borough pairs differed while controlling for multiple comparisons.

The Kruskal–Wallis test produced an H-statistic of 5070.54 with a p-value < 0.001. Since the p-value is less than the significance level of 0.05, the null hypothesis (H_{01}) was rejected, indicating that resolution times differ significantly across boroughs. The effect size (epsilon-squared = 0.0361) suggests a small to moderate effect of borough location on service resolution time. The analysis showed that Staten Island has the longest median resolution time, while Manhattan has the shortest.

The median resolution time was visualised across boroughs, showing clear differences in service performance. Staten Island has the highest median resolution time, while Manhattan has the lowest, supporting the statistical results *(See*

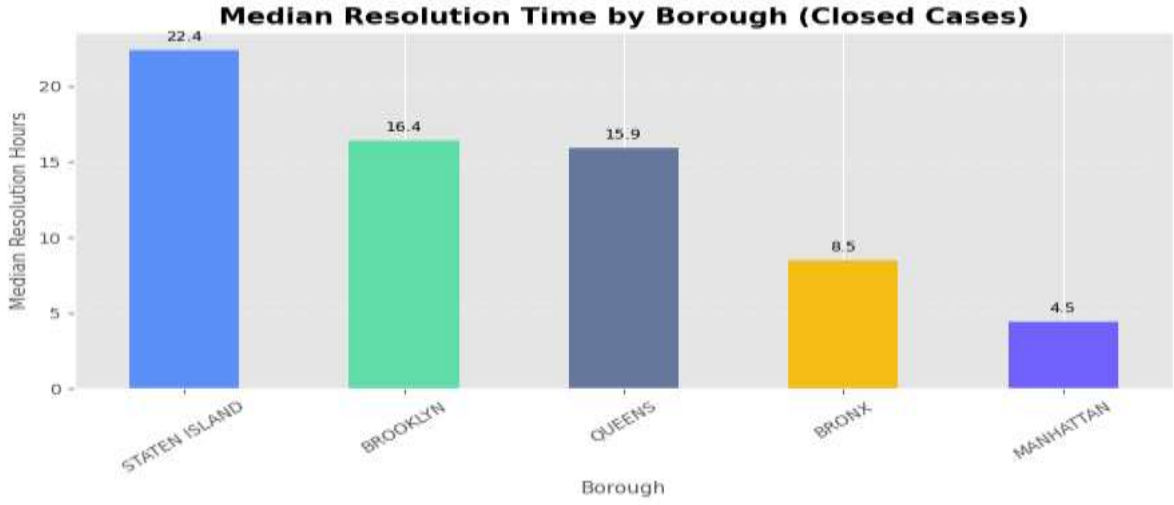

Figure 9: *Median Resolution Time by Borough (Closed Cases).*

3.1.2 DIAGNOSTICS AND VALIDATION

Several checks were conducted to ensure the validity of the analysis. The dataset was restricted to valid closed service requests with complete borough information. The distribution of resolution_hours was examined and found to be highly right-skewed, confirming the suitability of a non-parametric statistical test.

A log-scaled boxplot *(See **Figure 10**: Resolution Time by Borough (Closed Cases) — Log Scale)* was used to visualize resolution time distributions across boroughs, allowing both short and long resolution durations to be clearly represented. In addition, the Dunn post-hoc test confirmed that multiple borough pairs show statistically significant differences in resolution times.

These diagnostic results support the conclusion that service performance varies geographically across the boroughs of New York City.

3.2 HYPOTHESIS 2: AGENCY AND COMPLAINT TYPE ASSOCIATION WITH DELAY

3.2.1 METHODS APPLIED

Hypothesis 2 examined whether agency and complaint type are associated with the probability of delayed service resolution in the NYC 311 dataset. Two analytical approaches were applied: the Chi-Square Test of Independence and Logistic Regression Modelling.

The Chi-Square test was used to determine whether a statistical association exists between categorical variables and delayed resolution. Contingency tables were created using the binary delay variable (is_delayed) and the categorical predictors Agency and Complaint Type. For example, agency categories included DEP and DOT, while complaint types included Street Light Condition, Water System, and Sewer.

The Chi-Square tests produced significant results for both agency ($\chi^2 = 34{,}476.82$, $p < 0.001$) and complaint type ($\chi^2 = 34{,}728.18$, $p < 0.001$), indicating that delay probability differs across these categories. Effect size was measured using Cramér's V, which produced a value of approximately 0.49, indicating a strong association.

To support these findings, delay rates were visualised across categories. The results show that DOT has a higher delay rate compared to DEP, while Street Light Condition complaints exhibit the highest delay rate among complaint types *(See **Figure 11**: Delay Rate by Agency)*,*(See **Figure 12**: Delay Rate by Complaint Type).*

To further evaluate the predictive relationship, a logistic regression model was implemented to predict delay probability using Agency and Complaint Type as predictors. Categorical variables were converted into dummy variables using one-hot encoding, and the dataset was divided into 70% training data and 30% testing data using stratified sampling. Model coefficients were converted into odds ratios to interpret the effect of each category on delay probability.

3.2.2 DIAGNOSTICS AND VALIDATION

Several diagnostic checks were conducted to ensure the reliability of the analysis. For the Chi-Square test, expected cell frequencies were examined and confirmed to be greater than 5, satisfying the assumptions of the test.

The logistic regression model was evaluated using classification metrics, including confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC. The model achieved an accuracy of approximately 83% and a ROC-AUC score of approximately 0.775, indicating good predictive performance and performance substantially better than random classification.

The model performance is further illustrated using a confusion matrix, which shows the classification of delayed and non-delayed cases, and a ROC curve, which demonstrates the model's ability to distinguish between the two classes *(See **Figure 13**: Logistic Regression Confusion Matrix),(See **Figure 14**: Logistic Regression ROC Curve).*

Model interpretation showed that certain agencies and complaint categories significantly increase the likelihood of delayed resolution. For example, requests handled by DOT had higher odds of delay compared to DEP, while Street Light Condition complaints exhibited the highest delay probability, followed by Water System complaints. These results indicate that both agency assignment and complaint type play an important role in determining service delays.

Based on these results, the null hypothesis (H_{02}) was rejected, confirming that agency and complaint type are significantly associated with the probability of delayed service resolution in the NYC 311 system.

3.3 HYPOTHESIS 3: TEMPORAL FACTORS AND DELAY PREDICTION

3.3.1 METHODS APPLIED

Hypothesis 3 examined whether temporal factors influence the probability of delayed service resolution in NYC 311 service requests. The temporal variables analysed included weekend status (is_weekend), season (season), day of the week (created_weekday), and month of request (created_month).

Two analytical approaches were applied. First, Chi-Square tests of independence were conducted to examine whether delay probability differs across temporal categories such as weekday versus weekend and seasonal periods *(see **Figure 15**: Delay Rate Weekday vs Weekend,*

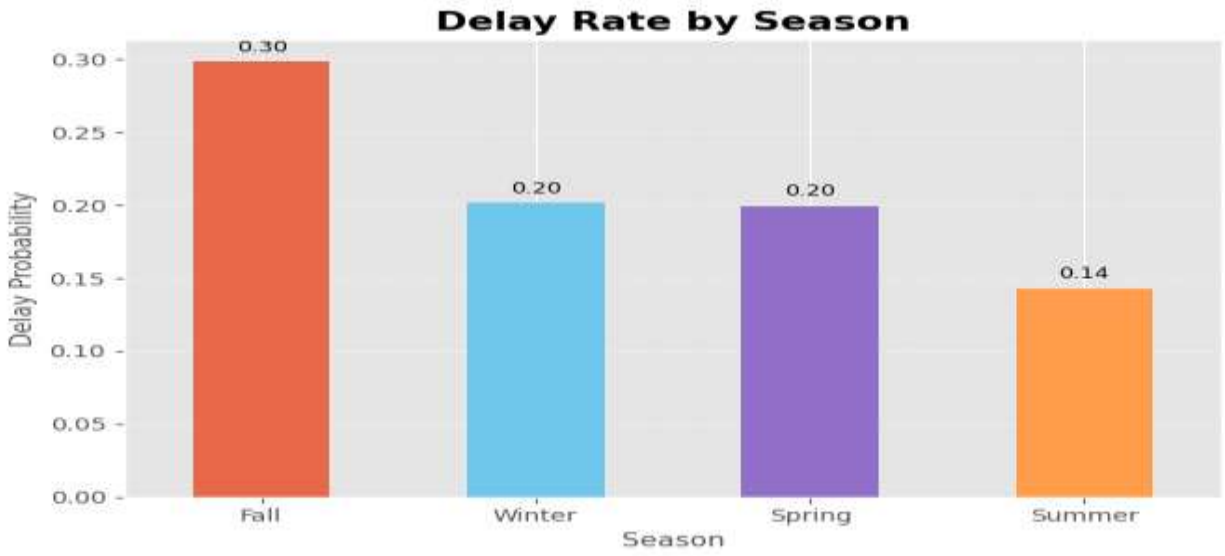

***Figure 16**: Delay Rate by Season)*. These tests evaluate whether there is a statistically significant association between categorical temporal variables and delayed service resolution.

Second, machine learning models were implemented to evaluate the predictive capability of temporal variables. A Logistic Regression model was used as a baseline classifier, while a Random Forest classifier was applied to capture nonlinear relationships and interactions among temporal features.

Before model training, categorical variables were converted into numerical form using one-hot encoding. The dataset was then divided into training (80%) and testing (20%) sets using stratified sampling to preserve the distribution of delayed and non-delayed service requests.

3.3.2 DIAGNOSTIC AND VALIDATION

Diagnostic checks were conducted to ensure the reliability of the statistical tests and machine learning models. For the Chi-Square analysis, contingency tables were examined and assumptions were verified, with Cramér's V used to measure effect size. Although temporal variables showed statistically significant relationships with delay probability, the strength of these associations was relatively weak.

Machine learning models were evaluated using confusion matrices, classification reports, and ROC curves. Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. During model evaluation, the logistic regression output generated a warning indicating that precision was undefined for one class because the model did not predict any instances of the delayed category in some cases. This occurred because the dataset is imbalanced, with non-delayed requests representing the majority class. As a result, although temporal factors showed statistically significant associations with delay probability, the logistic regression model had limited practical ability to identify delayed cases. This further supports the conclusion that temporal variables alone are weak predictors compared with operational features such as complaint type and agency workload. The Logistic Regression model achieved a ROC-AUC of 0.598(*See **Figure 17**: H3 Logistic Regression ROC Curve)*, while the Random Forest model slightly improved performance with a ROC-AUC of 0.646*(See **Figure 18**: H3 Random Forest ROC Curve)*. Feature importance analysis *(See*

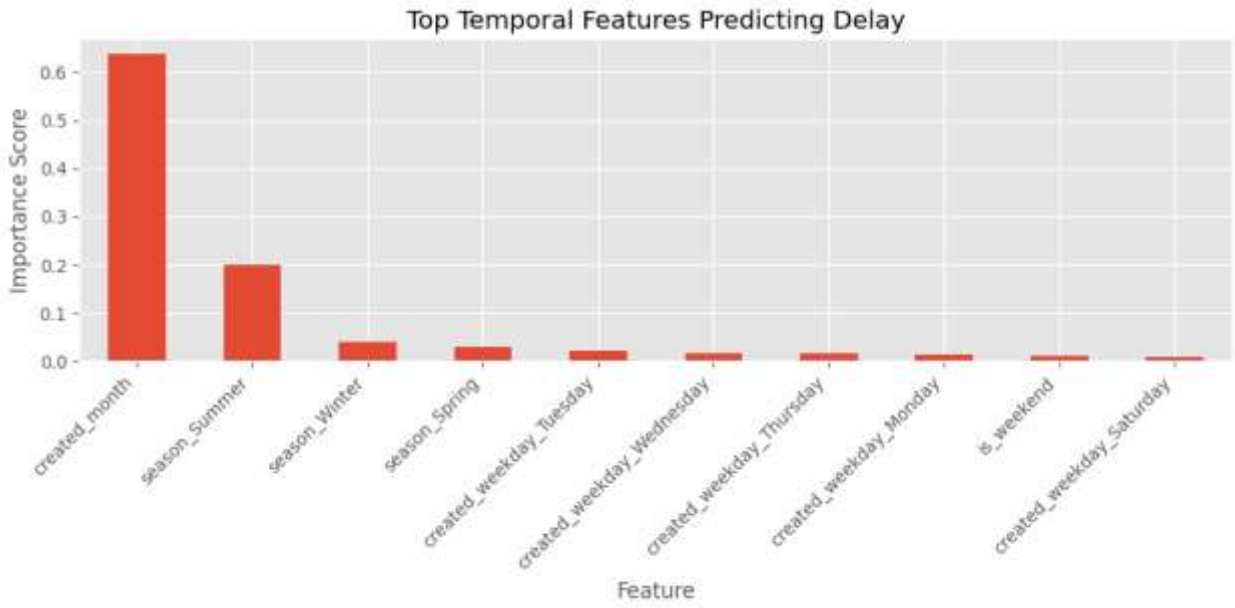

***Figure 19**: Top Temporal Features Predictions Delay)* indicated that created_month had the strongest influence on delay prediction, while the weekend indicator had minimal impact.

Overall, the results suggest that temporal variables have limited predictive power, and operational factors such as agency workload and complaint type are likely stronger predictors of service delays**.**

3.4 HYPOTHESIS 4: DELAY PREDICTION USING HISTORICAL SERVICE REQUEST DATA

3.4.1 METHODS APPLIED

Hypothesis 4 examined whether historical service request attributes can significantly predict the delayed resolution of NYC 311 service requests. To test the hypothesis, a multivariable logistic regression model was implemented in Python. The dependent variable was is_delayed, a binary variable indicating whether a service request experienced delayed resolution.

The independent variables used in the model included agency workload (agency_workload_daily), borough, complaint type, seasonal indicator (season), and weekend indicator (is_weekend). Categorical variables such as borough and complaint type were incorporated using dummy variable encoding. To evaluate predictive performance realistically, a time-based train–test split was used, where the earliest 70% of observations were used for model training, and the remaining 30% were used for model testing.

The logistic regression coefficients were examined to understand the influence of each predictor on delayed service resolution. A feature importance visualisation based on regression coefficients was also produced to highlight the most influential predictors *(See*

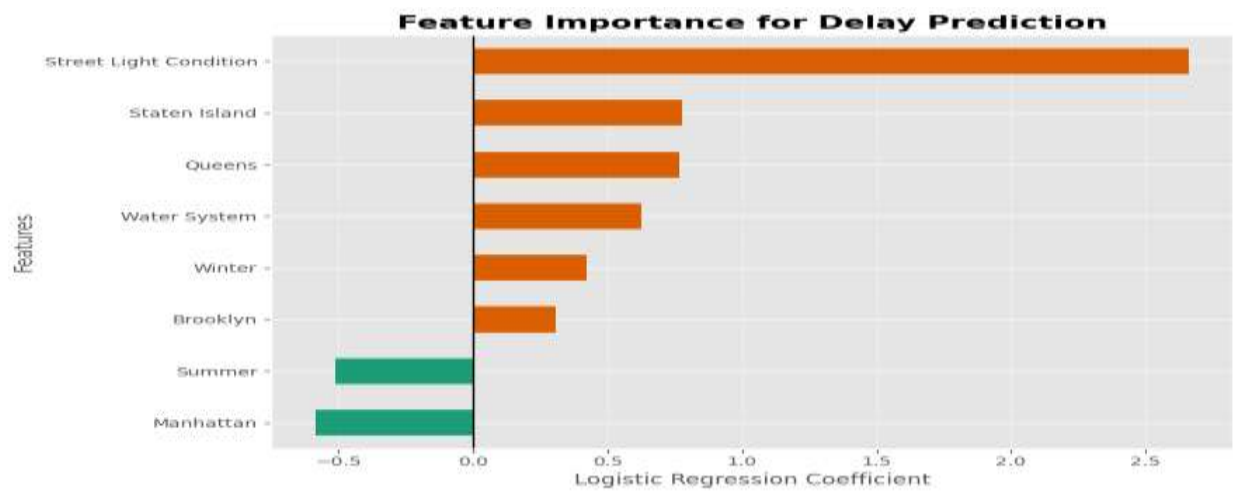

Figure 20: *Logistic Regression Feature Importance).*

To further strengthen the analysis, an additional machine learning method, Decision Tree Classification, was implemented. The decision tree model used controlled parameters (max_depth, min_samples_split, and min_samples_leaf) to prevent overfitting and improve generalisation. Feature importance scores from the decision tree were also analysed to identify key operational predictors influencing service delays (*See*

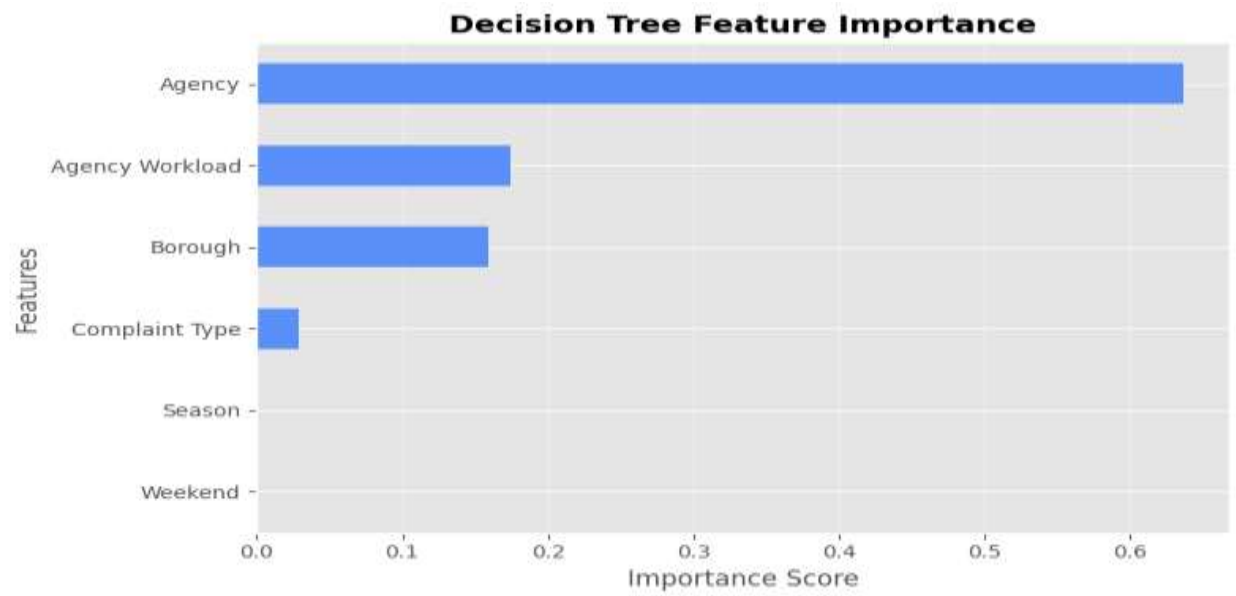

Figure 21: *Decision Tree Feature Importance).*

3.4.2 DIAGNOSTICS AND VALIDATION

Several diagnostic checks were conducted to ensure the reliability and validity of the predictive models. Multicollinearity among predictor variables was assessed using the Variance Inflation Factor (VIF), and the results indicated acceptable values, confirming that multicollinearity was not a concern. A correlation heatmap was also generated to examine relationships among key variables, showing a moderate positive relationship between resolution hours and delayed cases, while other variables, such as weekend indicators, showed weak correlations.

Model interpretability was evaluated using logistic regression coefficients and decision tree feature importance. The results indicated that complaint categories such as Street Light Condition and Water System, along with operational factors like agency workload and borough, significantly influenced delayed service resolution.

Model performance was evaluated using confusion matrices *(See*

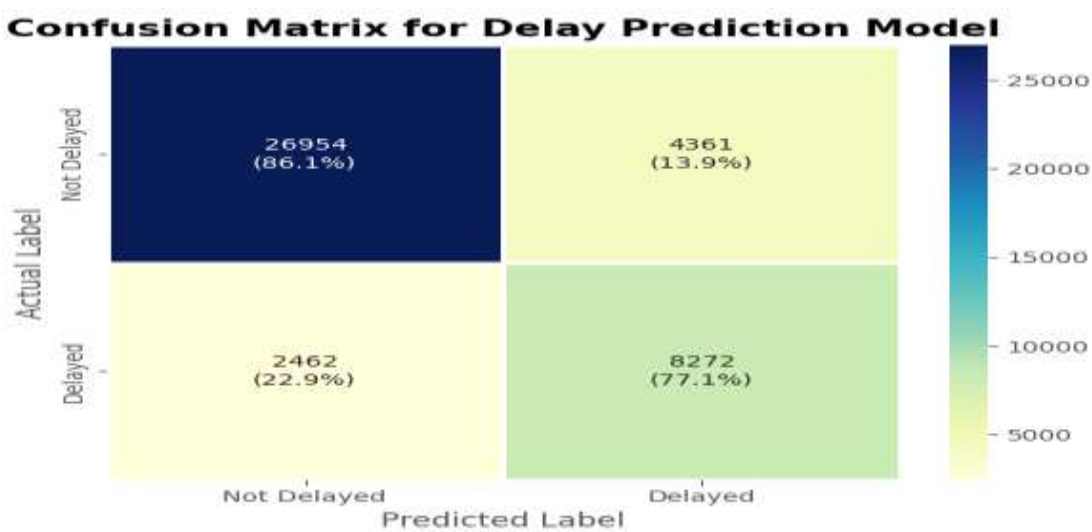

***Figure 22**: Confusion Matrix for the Delay Prediction model,*

***Figure 23**: Decision Tree Confusion Matrix)*, classification reports, and ROC curves *(See*

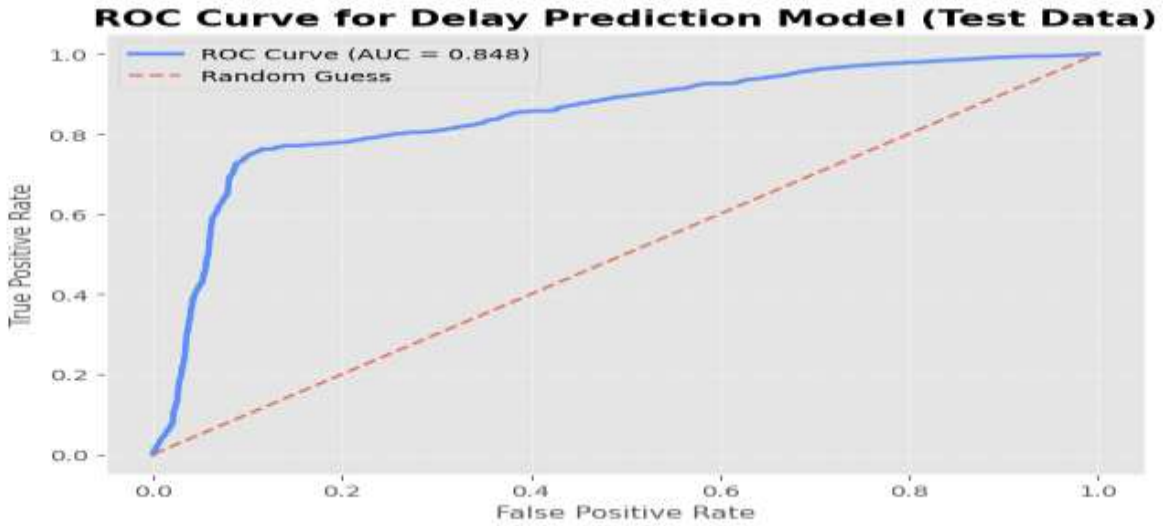

***Figure 24**: ROC Curve for Delay Prediction Model, **Figure 25**: Model Calibration Plot)*. The logistic regression model achieved an AUC of approximately 0.85, while the decision tree model slightly outperformed it with an AUC of approximately 0.86. Overall, both models demonstrated strong predictive performance. Since the AUC values exceeded 0.70, the null hypothesis was rejected, indicating that historical service request attributes significantly predict delayed resolution

in NYC 311 service requests. To compare the performance of different predictive models, a summary of key evaluation metrics is presented in (*See* ***Table 2*****:** Summary of Predictive Model Performance *Summary of Predictive Model Performance)*.

3.5 HYPOTHESIS 5: AGENCY WORKLOAD AND DELAY PROBABILITY

3.5.1 METHODS APPLIED

Hypothesis 5 examined whether agency workload significantly influences the probability of delayed resolution for NYC 311 service requests.

To test the hypothesis, a **simple logistic regression model** was implemented using Python. The dependent variable was **is_delayed**, a binary indicator representing whether a service request experienced delayed resolution.

The independent variable used in the model was **daily agency workload (agency_workload_daily)**, which represents the number of service requests handled by the agency on a given day.

The logistic regression model estimates how changes in agency workload affect the probability that a service request will be delayed. Model coefficients were interpreted using **odds ratios** to quantify the impact of workload increases on delay probability.

A **boxplot visualization** *(See*

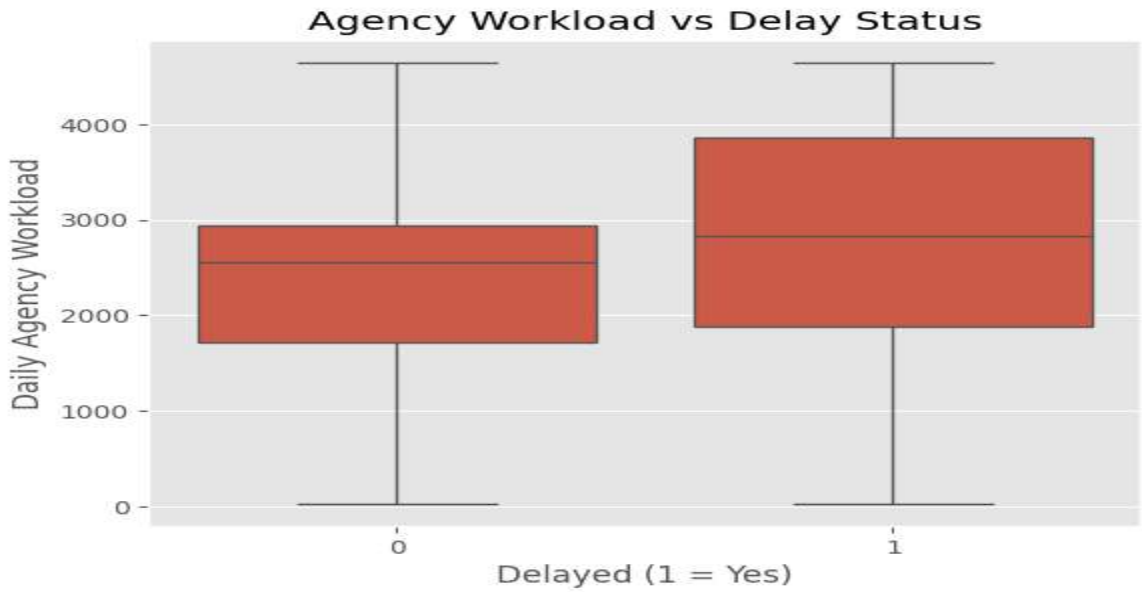

***Figure 26**: Agency Workload vs Delay Boxplot)* was also generated to compare the distribution of agency workload between delayed and non-delayed service requests.

3.5.2 DIAGNOSTICS AND VALIDATION

The regression model converged successfully, indicating stable parameter estimation. The coefficient for agency_workload_daily was positive and statistically significant, suggesting that higher workload is associated with a greater likelihood of service delays.

To interpret the practical impact of workload changes, the coefficient was transformed into an odds ratio for a 100-unit increase in workload. The results indicated that a 100-unit increase in agency workload increases the odds of delayed service resolution by approximately 4.3%.

The boxplot visualization comparing agency workload and delay status supports this finding by showing that delayed cases tend to occur at higher levels of daily agency workload.

The statistical significance of the logistic regression coefficient was assessed using the p-value from the regression model. The p-value for the agency workload variable was less than 0.05, indicating that the relationship between workload and delay probability is statistically significant.

Because the regression coefficient is positive and statistically significant, the null hypothesis was rejected. Higher agency workload significantly increases the probability of delayed resolution in NYC 311 service requests.

4 CHAPTER 4: DATA VISUALIZATION

4.1 VISUAL REPRESENTATION

To support the statistical analysis and make the findings easier to interpret, a Power BI dashboard was developed using six interactive report pages. These dashboards summarise service volume, resolution time, delay patterns, geographic distribution, operational trends, and detailed comparisons across complaint types and boroughs.

4.1.1 DASHBOARD OVERVIEW

The dashboard overview presents the main performance indicators of the NYC 311 service request system. The visuals show that the dataset contains approximately 149K service requests, with an average resolution time of 142.82 hours and an average agency workload of 2.52K. The overall delay rate is 20%, while 94.09% of requests were closed, indicating strong completion rates but also highlighting that delays remain a meaningful operational issue. *(See*

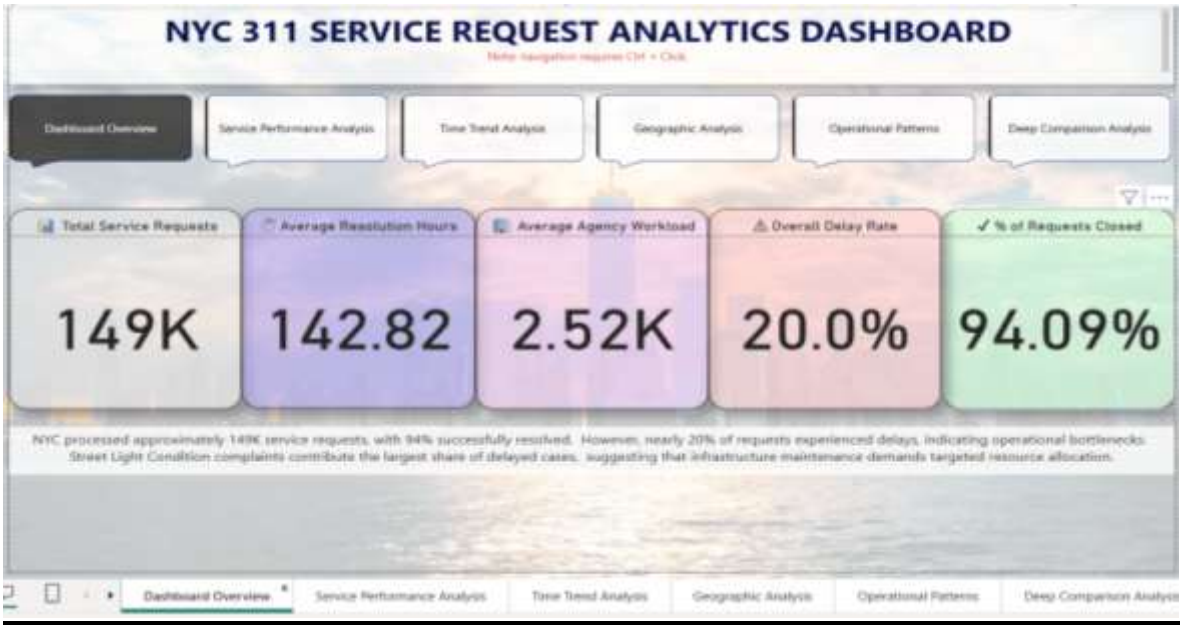

***Figure 27**: NYC 311 Service Request Analytics Dashboard Overview).*

4.1.2 SERVICE PERFORMANCE ANALYSIS

The service performance dashboard highlights differences across boroughs, agencies, complaint types, and seasons. Queens shows the highest average resolution time, followed by Brooklyn and Staten Island, while the Bronx has the lowest among the displayed boroughs. The delay rate by complaint type indicates that Street Light Condition complaints have the highest delay rate (56.77%), far above Water System (10.35%) and Sewer (5.92%). Agency-level analysis also shows that DOT experiences substantially higher delay rates than DEP, suggesting uneven operational pressure across agencies *(See **Figure 28**: Service Performance Analysis Dashboard).*

4.1.3 TIME TREND ANALYSIS

The time trend dashboard shows that service demand and delays vary over time. Request volumes fluctuate across months, while the **monthly delay rate increases sharply toward year-end**, reaching **42.99% in December**. Average resolution time declines during mid-year and rises again in later months, suggesting that operational performance changes across seasons rather than

remaining stable throughout the year *(See*

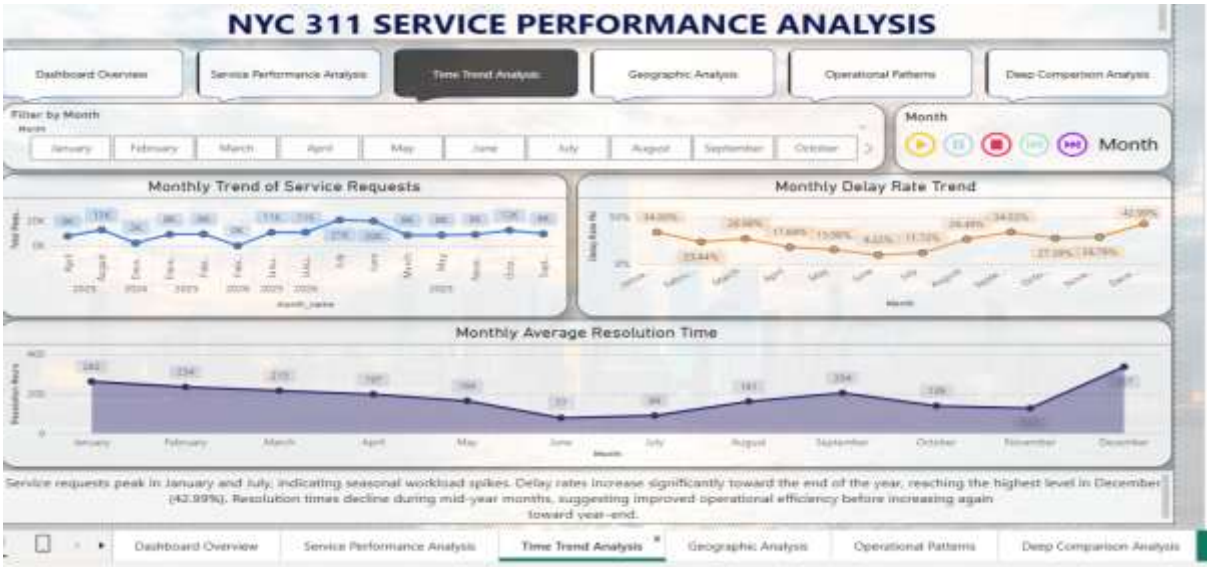

Figure 29*: Time Trend Analysis Dashboard).*

4.1.4 GEOGRAPHIC ANALYSIS

The geographic dashboard shows that complaint activity is concentrated in dense urban areas. The borough-level request volume confirms that Queens (41.85K) and Brooklyn (37.77K) account for the largest number of service requests, followed by the Bronx (30.56K), Manhattan (22.09K), and Staten Island (13.88K). This pattern suggests that service demand is strongly linked to borough size, infrastructure use, and population concentration *(See*

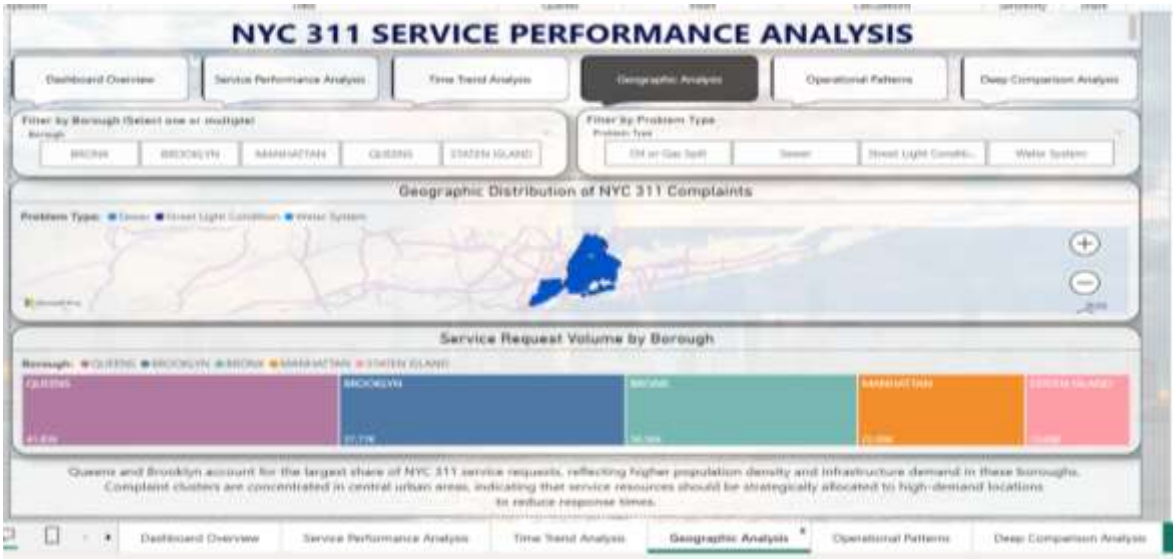

Figure 30: *Geographic Analysis Dashboard)*

4.1.5 OPERATIONAL PATTERNS

The operational dashboard focuses on weekday behavior. Service requests are highest on Tuesday and Wednesday, while weekend activity drops noticeably. The average delay rate is highest on Monday (21.92%) and Friday (21.68%), suggesting increased pressure at the beginning and end of the workweek. Complaint resolution time is also longest on Friday, indicating that operational backlog may build before the weekend *(See*

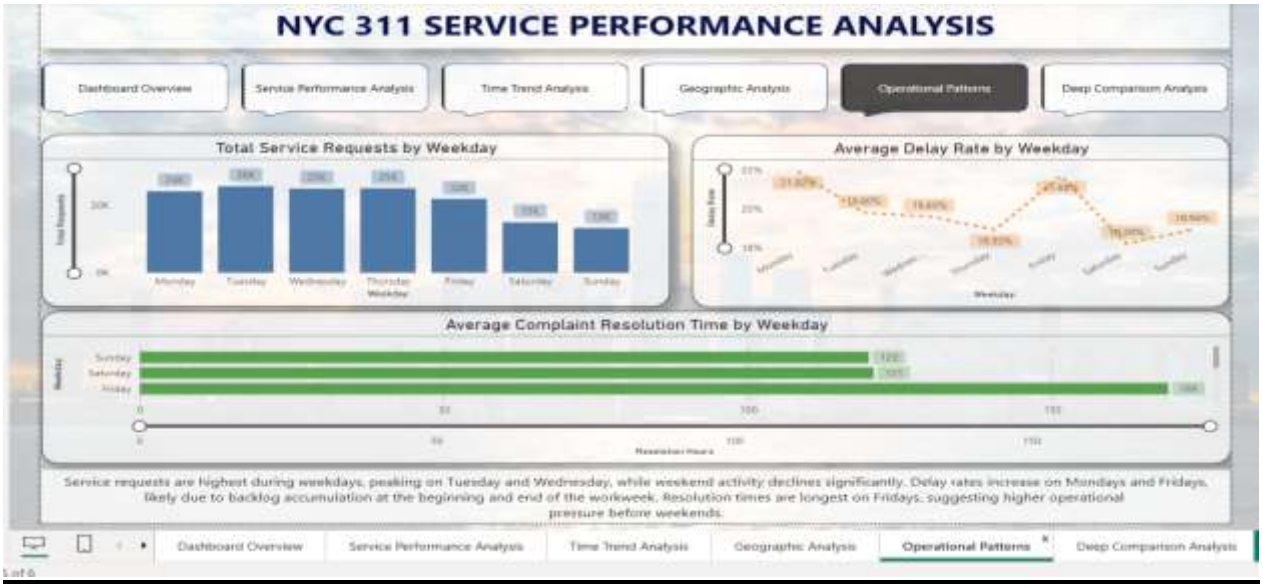

Figure 31: *Operational Patterns Dashboard).*

4.1.6 DEEP COMPARISON ANALYSIS

The deep comparison dashboard provides a more detailed cross-sectional view of complaint behavior. The monthly complaint table shows that Street Light Condition complaints dominate across most months, while Water System and Sewer issues vary at lower levels. The borough comparison also shows that Street Light Condition has the largest share of complaints in several boroughs, especially Staten Island (84.33%) and Brooklyn (70.30%). This dashboard confirms that infrastructure-related complaints are the main contributor to service delays and should be prioritized in planning and resource allocation *(See*

***Figure 32**: Deep Comparison Analysis Dashboard).*

4.1.7 SUMMARY OF VISUAL FINDINGS

Overall, the dashboards show that NYC 311 service performance varies across **boroughs, agencies, complaint types, months, and weekdays**. Delays are most severe for **Street Light Condition complaints**, are more common under **DOT**, and increase during specific **months and weekdays**. Geographic concentration in **Queens and Brooklyn** further suggests that service demand is unevenly distributed across the city. These visual findings support the statistical results and reinforce the need for targeted operational planning, better workload management, and proactive infrastructure maintenance.

5 CHAPTER 5: FINDINGS AND DISCUSSION

The analysis of the NYC 311 service request dataset produced several important findings regarding service resolution performance, delay patterns, and operational workload.

First, the results show that **resolution time differs significantly across boroughs**. The statistical analysis confirmed that geographic location plays a role in determining how quickly

service requests are resolved. Boroughs with larger populations and higher service demand tend to experience longer resolution times.

Second, the analysis demonstrated that **agency assignment and complaint type significantly influence delay probability**. Complaints handled by certain agencies, particularly those related to infrastructure maintenance such as street lighting, show a higher likelihood of delayed resolution. This indicates that operational workload and task complexity differ across agencies.

Third, **temporal factors also influence service performance**. The results reveal that both monthly and weekday patterns affect service request volume and delay rates. Service demand fluctuates throughout the year and across weekdays, which suggests that operational planning should account for seasonal and weekly workload variations.

Fourth, predictive modeling showed that **historical service request attributes can successfully predict delayed resolution**. Logistic regression analysis demonstrated that factors such as agency workload, borough, complaint type, and temporal features contribute significantly to predicting whether a service request will experience delays.

Finally, the results confirmed that **agency workload is positively associated with delay probability**. As agency workload increases, the likelihood of service delays also increases. This finding highlights the importance of effective workload distribution and resource management across municipal agencies.

Overall, these findings indicate that delays in NYC 311 service requests are influenced by a combination of **geographic, operational, temporal, and workload-related factors**.

The findings provide important insights into the operational dynamics of the NYC 311 service request system. The significant differences in resolution time across boroughs suggest that

geographic and demographic factors influence service demand and operational efficiency. Boroughs with higher population density and infrastructure complexity may experience greater service demand, leading to longer resolution times.

The strong relationship between complaint type and delay probability highlights the importance of **service category characteristics**. Infrastructure-related complaints, such as street light maintenance, often require physical inspections, equipment, or coordination between departments, which may increase resolution time.

Temporal patterns identified in the analysis indicate that service requests and delays are not evenly distributed across time. Certain months and weekdays show higher service demand and delay rates, suggesting that agencies may benefit from adjusting staffing levels or resource allocation during peak periods.

The predictive modeling results further demonstrate that historical service data can be used to anticipate service delays. By identifying high-risk cases earlier, municipal agencies could implement proactive strategies to reduce service delays and improve operational performance.

Finally, the relationship between agency workload and delay probability emphasizes the need for **efficient workload management**. Agencies handling a large volume of service requests may require additional resources or improved operational processes to maintain service efficiency.

6 CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 CONCLUSION

This study analysed NYC 311 service request data to investigate factors influencing service resolution time and delay probability. Using statistical analysis, predictive modelling, and

interactive visualisations, the study identified several key operational patterns within the municipal service system.

The results show that service resolution performance varies significantly across boroughs, agencies, complaint types, and time periods. Infrastructure-related complaints and high agency workload were found to be major contributors to delayed service resolution. Additionally, temporal patterns in service demand suggest that workload pressure fluctuates throughout the year and across weekdays.

The findings demonstrate that combining statistical analysis with data visualisation can provide valuable insights into urban service operations. These insights can support better decision-making and improve the efficiency of municipal service management systems.

6.2 RECOMMENDATIONS

Based on the findings of this study, several recommendations can be made to improve NYC 311 service performance:

1. **Improve Resource Allocation Across Boroughs**

   Municipal agencies should allocate resources based on service demand patterns across boroughs. Boroughs with higher request volumes may require additional operational support.

2. **Prioritise High-Delay Complaint Categories**

   Infrastructure-related complaints, such as street light issues, should receive priority attention, as they exhibit the highest delay rates.

3. **Implement Predictive Monitoring Systems**

   Predictive models developed in this study could be integrated into operational systems

to identify service requests with a high probability of delay and enable proactive intervention.

4. **Optimise Agency Workload Distribution**

   Balancing workloads across agencies may help reduce operational pressure and improve resolution efficiency.

5. **Use Temporal Planning for Staffing and Operations**

   Agencies should consider seasonal and weekday demand patterns when scheduling staff and allocating operational resources.

By implementing these strategies, municipal agencies can improve response efficiency, reduce service delays, and enhance the overall effectiveness of public service delivery.

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APPENDICES

1. APPENDIX A: CLEANED DATASET

The datasets, metadata file, Python scripts, and Power BI dashboard developed in this study are available through the following Google Drive repository:

Google Drive Link:

The repository includes the following files:

- Raw dataset (311_Service_Requests.csv)
- Cleaned dataset (311_clean_base.csv)
- Metadata file (311_Service_Requestes_Meta_Data.xlsx)

2. APPENDIX B: PYTHON ANALYTICAL SCRIPTS

The Python code files used in this study are available through the following Google Drive repository.

Google Drive Link: Python Script

The repository includes the Python scripts developed for data preprocessing, exploratory data analysis, feature engineering, statistical testing, predictive modeling, and dashboard preparation.

In addition, an HTML export of the notebook is provided, which contains both the code and its corresponding outputs. This file is included to improve transparency, reproducibility, and ease of review.

3. APPENDIX C: TABLE

Table Name	Purpose	Attributes
NYC 311 Service Requests	Stores records of non-emergency municipal service requests submitted by residents	Unique Key, Created Date, Closed Date, Agency, Borough, Complaint Type, Descriptor,Status, Latitude, Longitude

Table 1: *Description of NYC 311 Service Request Dataset Attributes*

Model	AUC
Logistic Regression	0.85
Decision Tree	0.86
Random Forest	>0.85

Table 2: *Summary of Predictive Model Performance*

4. APPENDIX D: VISUALISATION

Figure 1: *Boxplot of Resolution Time Distribution*

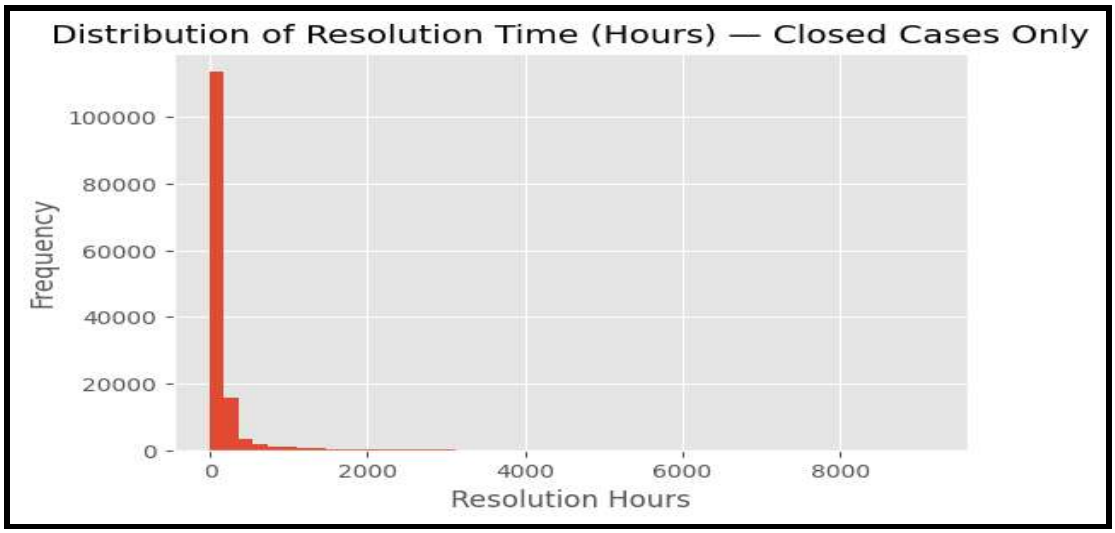

Figure 2: *Histogram of Resolution Time Distribution*

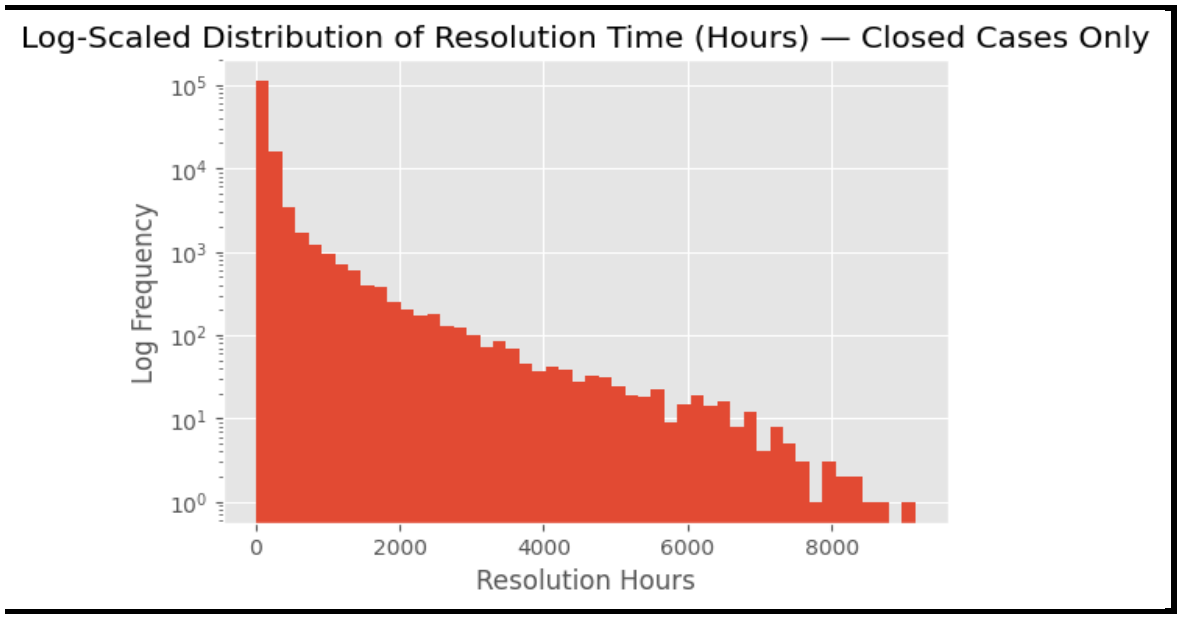

Figure 3: *Log-Scaled Distribution of Resolution Time (Hours) — Closed Cases Only*

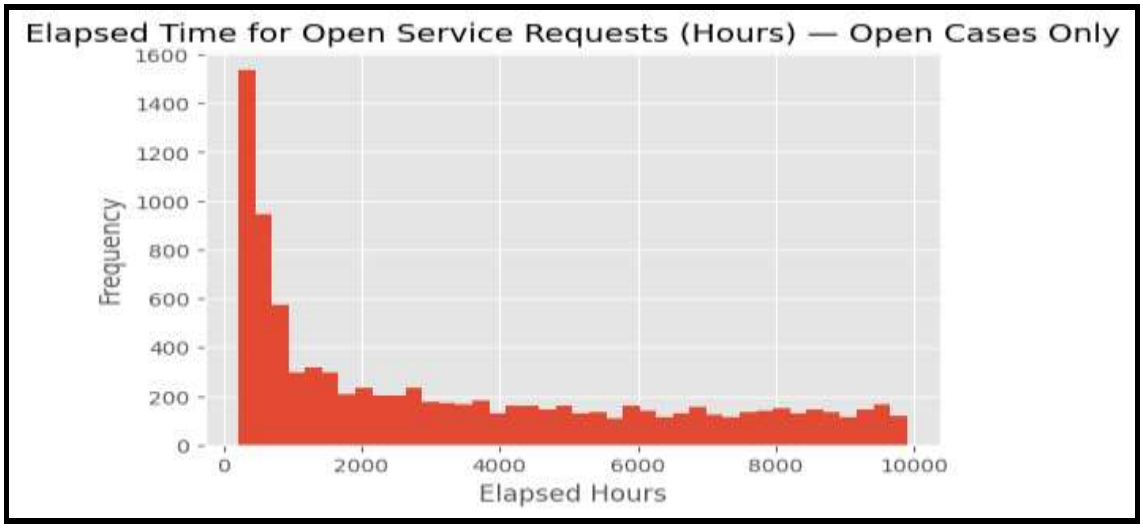

Figure 4: *Elapsed Time for Open Service Requests*

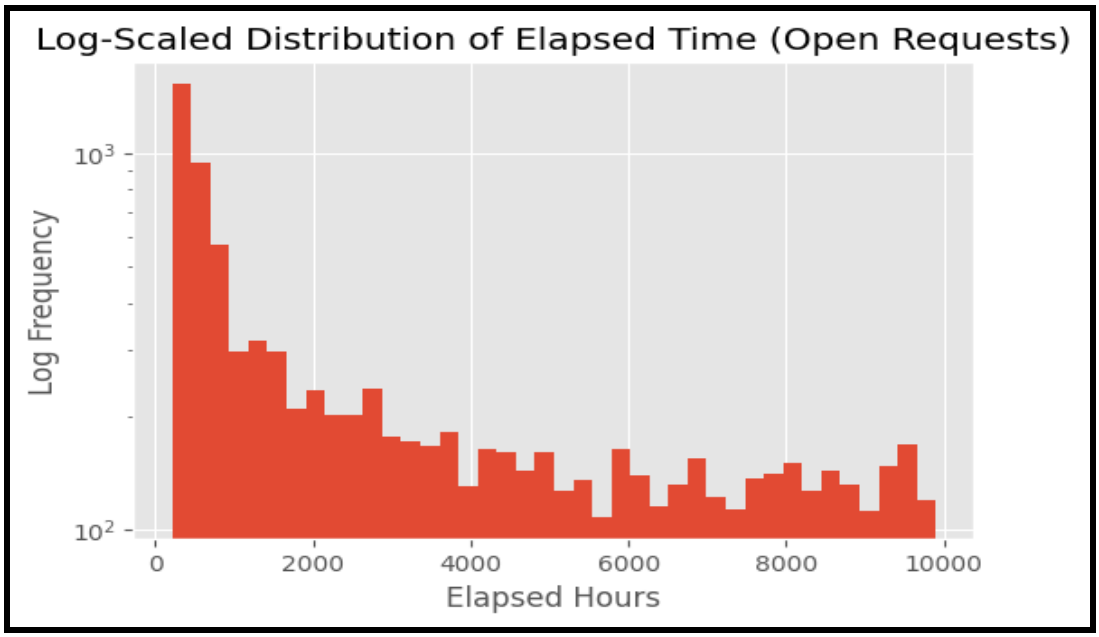

Figure 5: *Log Elapsed Time for Open Service Requests*

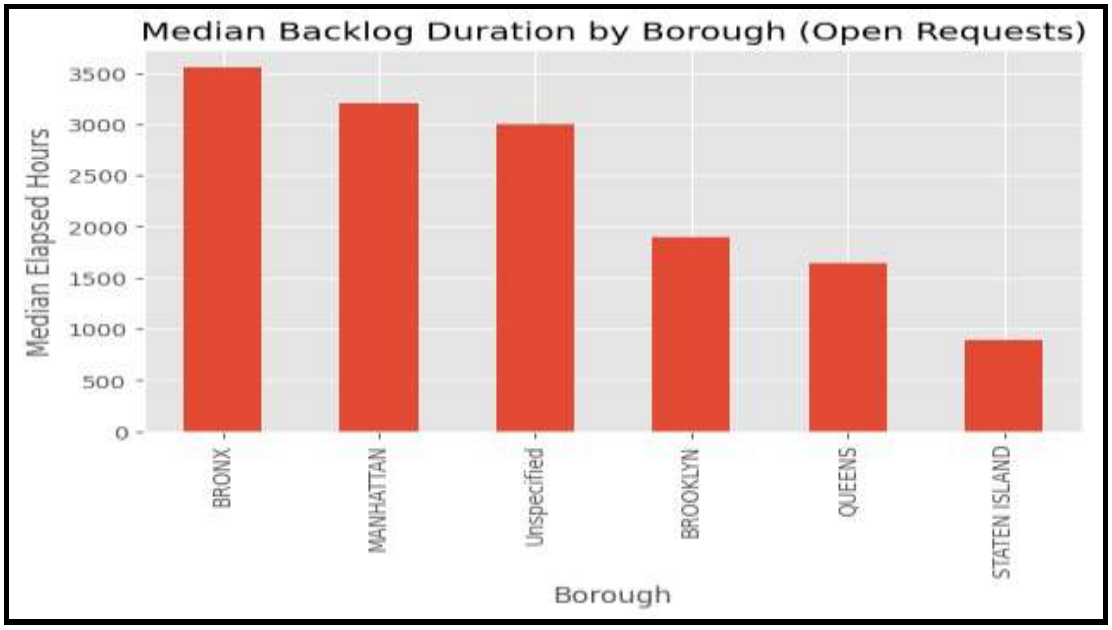

Figure 6: *Median Backlog Duration*

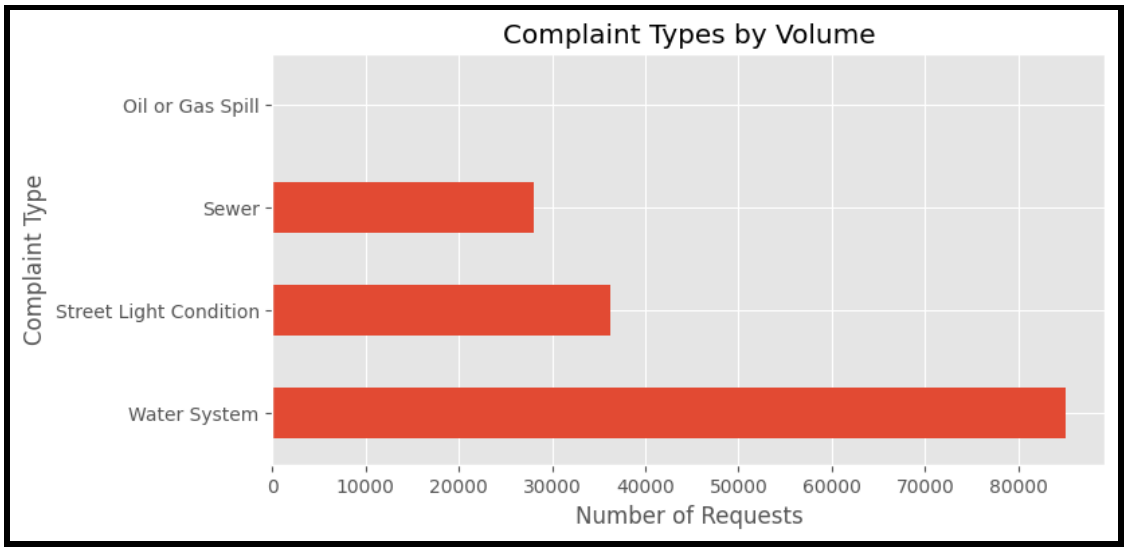

Figure 7: *Complaint Types by Volume*

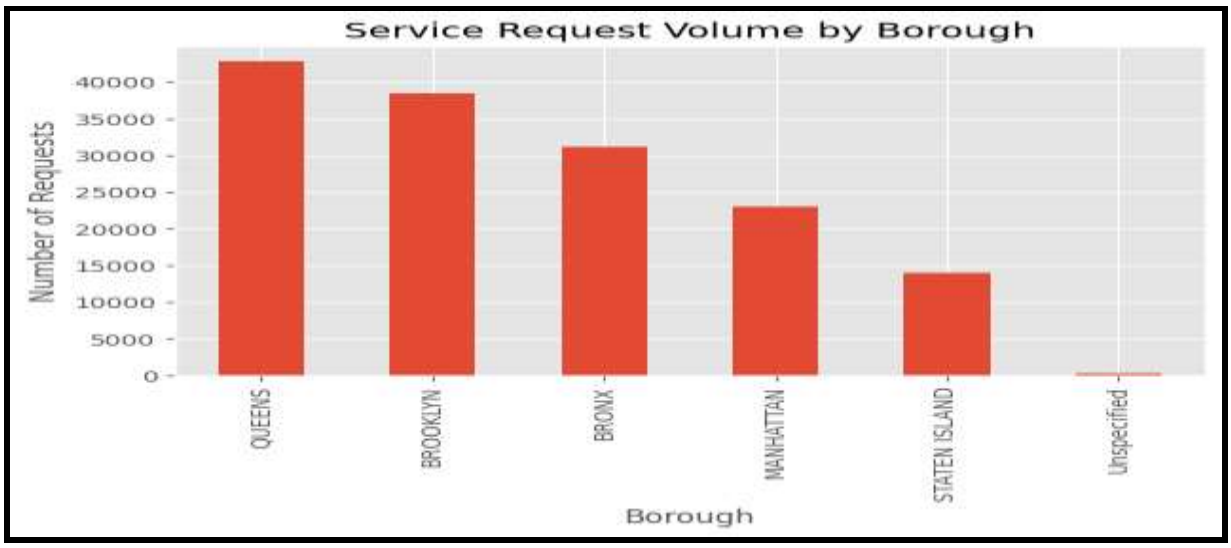

Figure 8: *Service Requests by Borough*

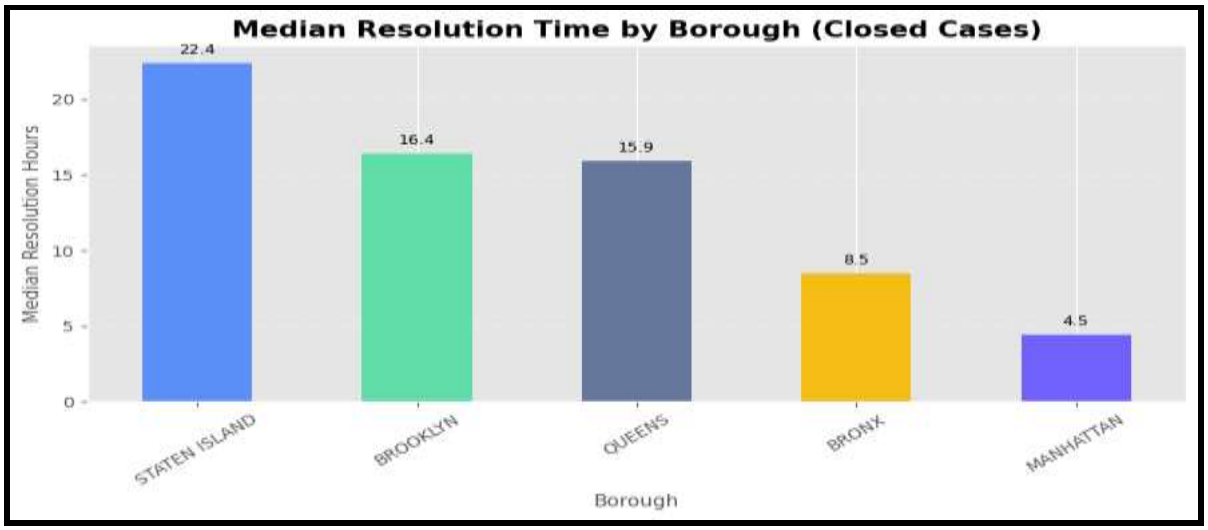

Figure 9: *Median Resolution Time by Borough (Closed Cases)*

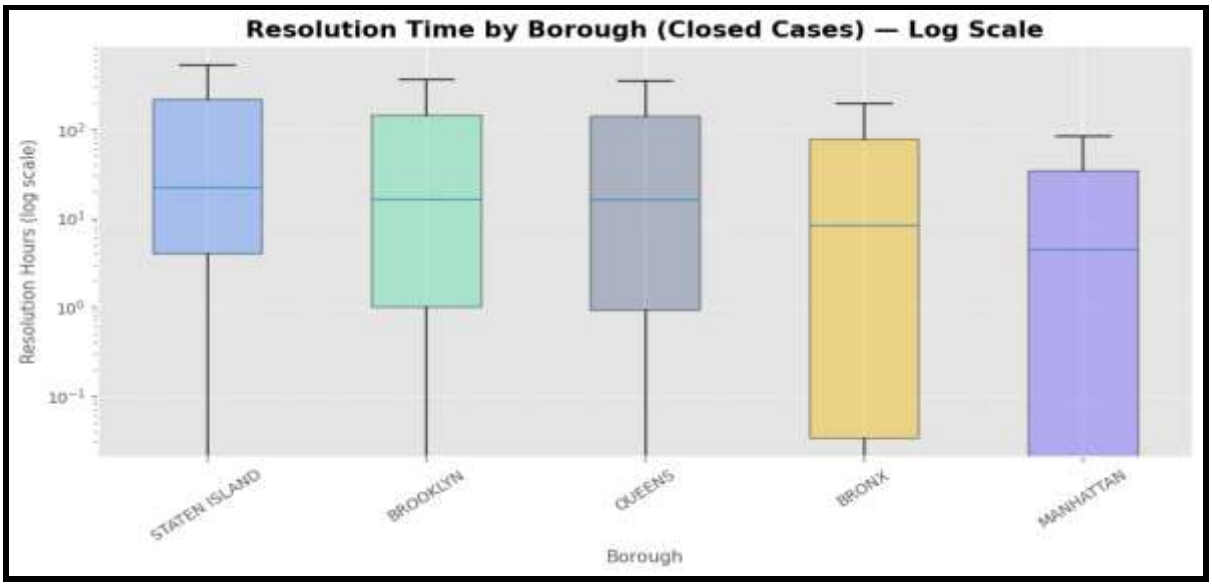

Figure 10: *Resolution Time by Borough (Closed Cases) — Log Scale*

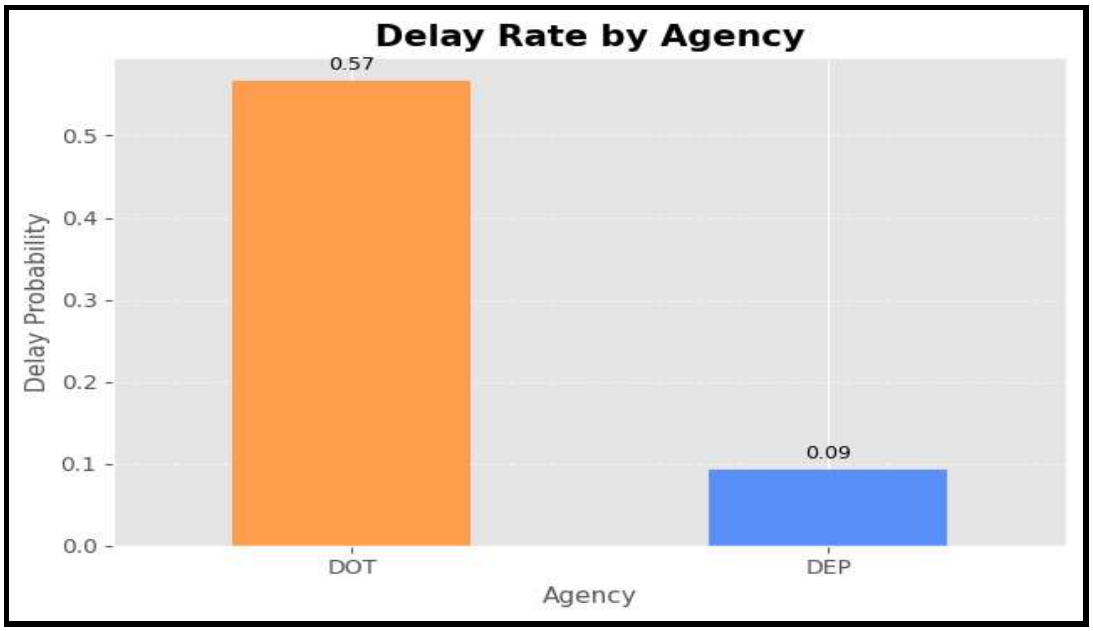

Figure 11: *Delay Rate by Agency*

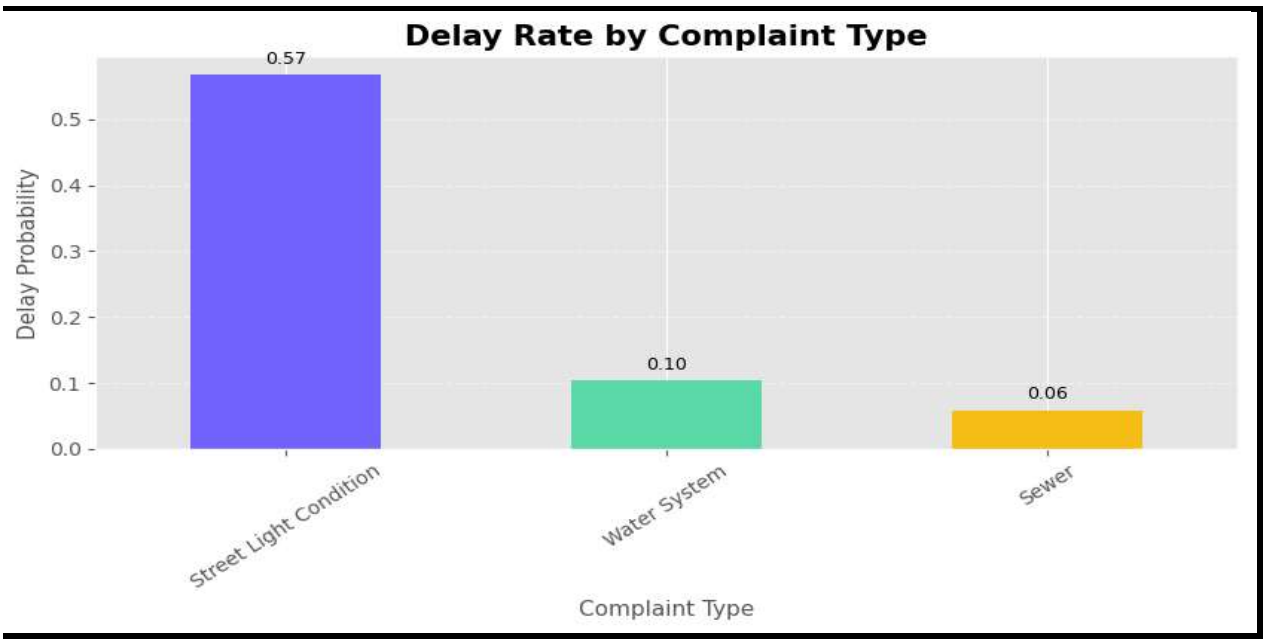

Figure 12: *Delay Rate by Complaint Type*

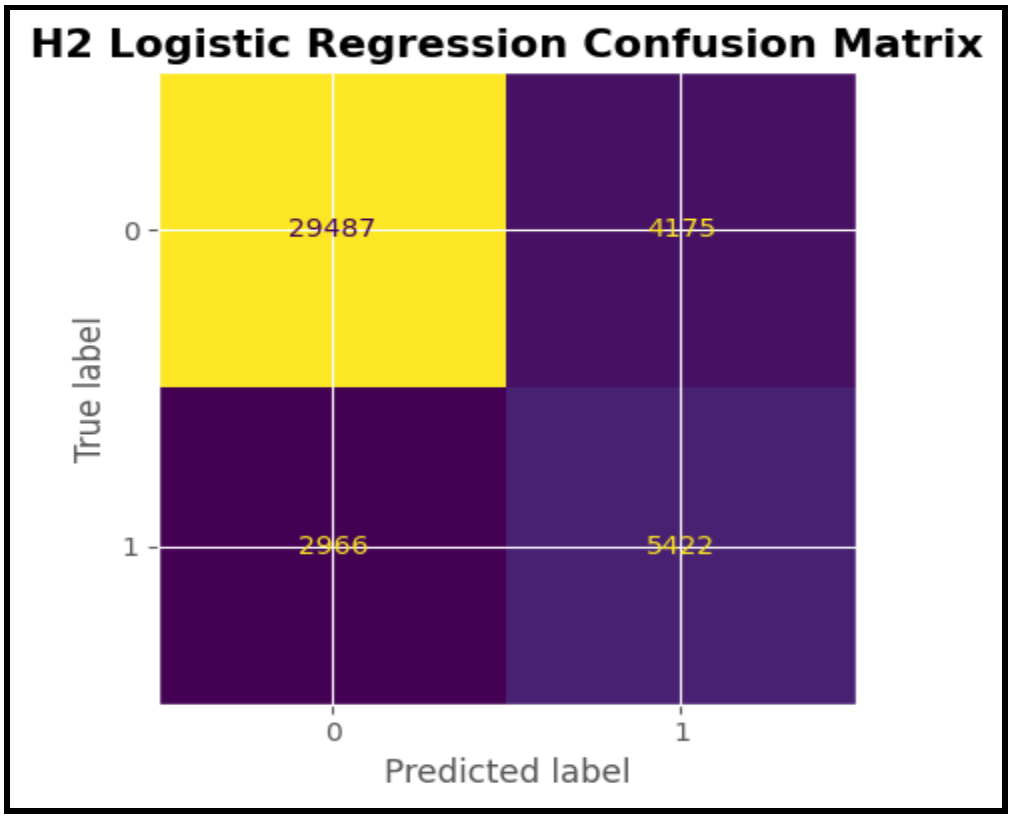

Figure 13: *Logistic Regression Confusion Matrix*

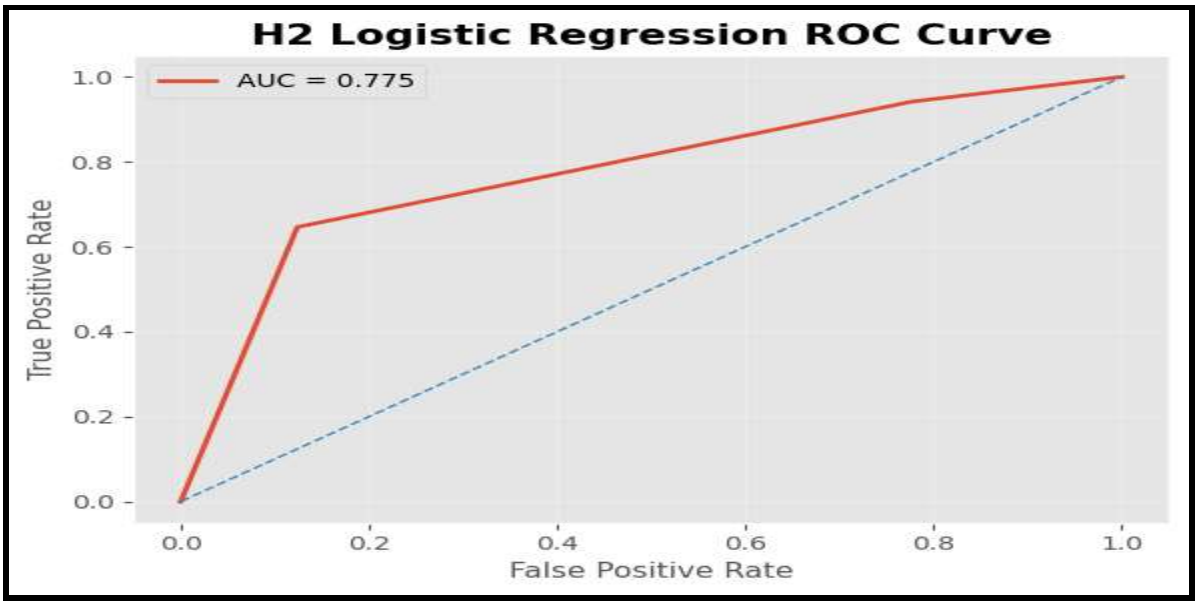

Figure 14: *Logistic Regression ROC Curve*

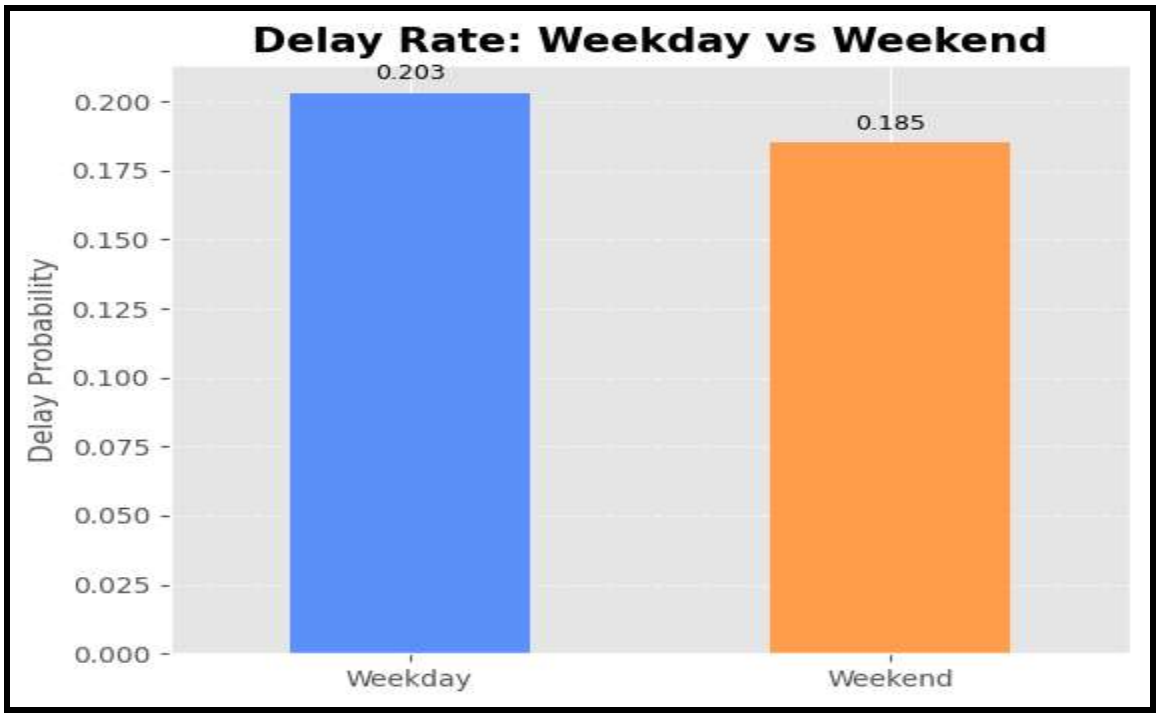

Figure 15: *Delay Rate Weekday vs Weekend*

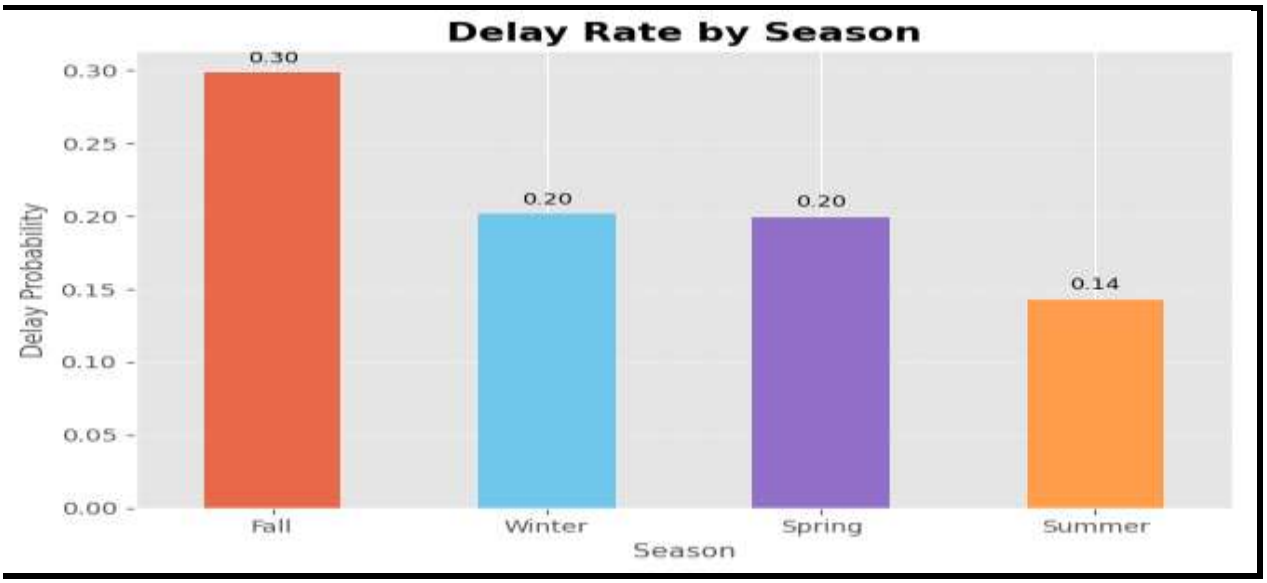

Figure 16: *Delay Rate by Season*

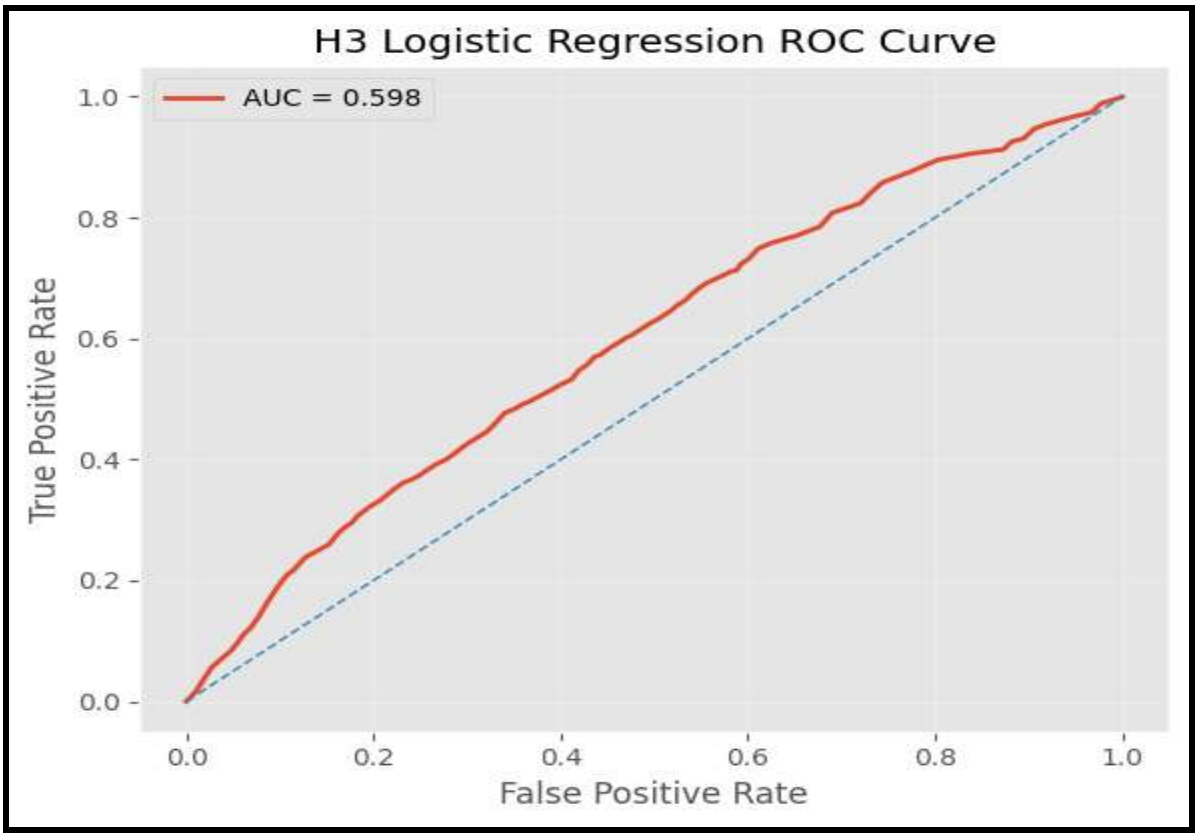

Figure 17: *H3 Logistic Regression ROC Curve*

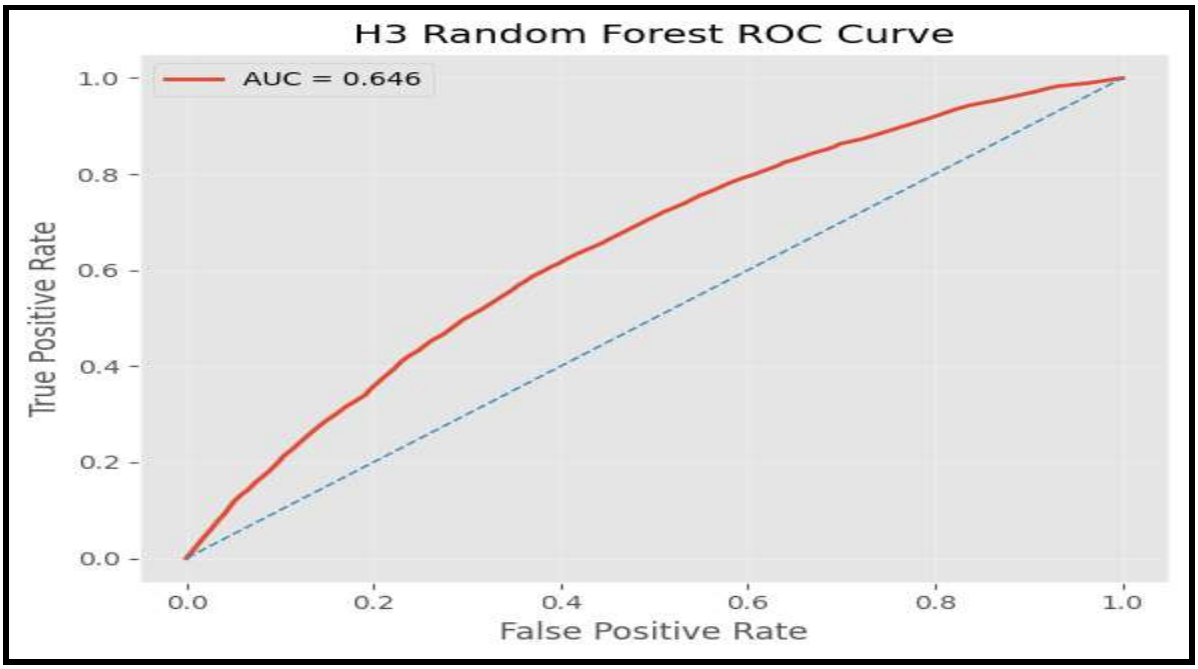

Figure 18: *H3 Random Forest ROC Curve*

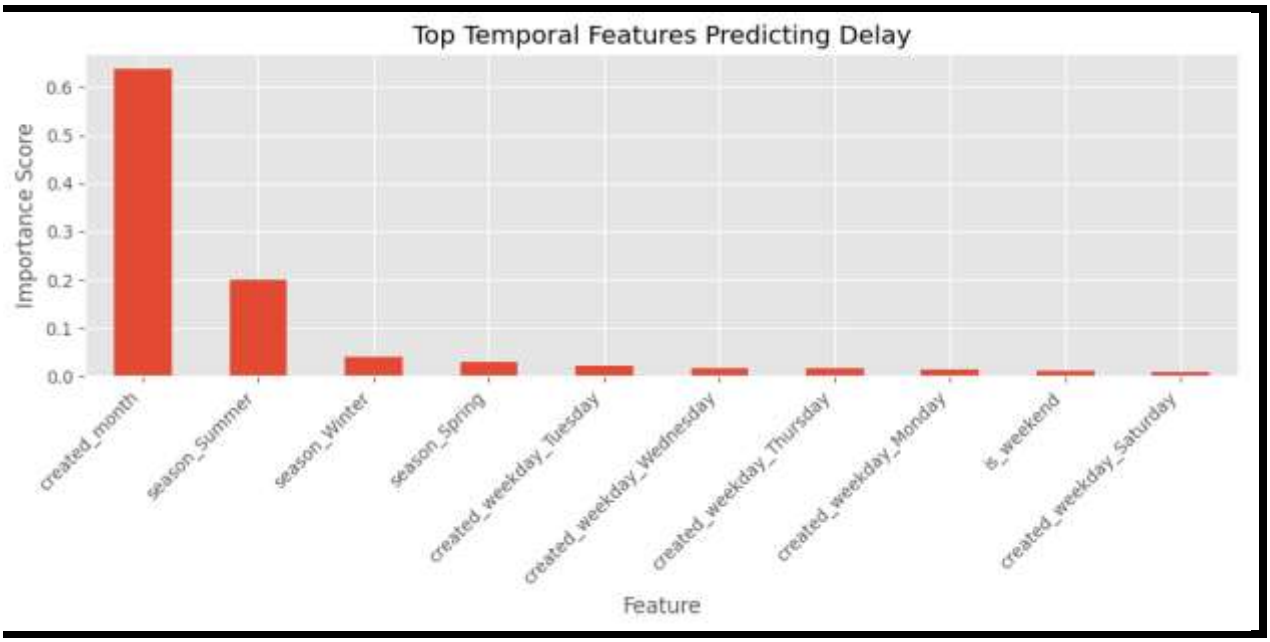

Figure 19: *Top Temporal Features Predictions Delay*

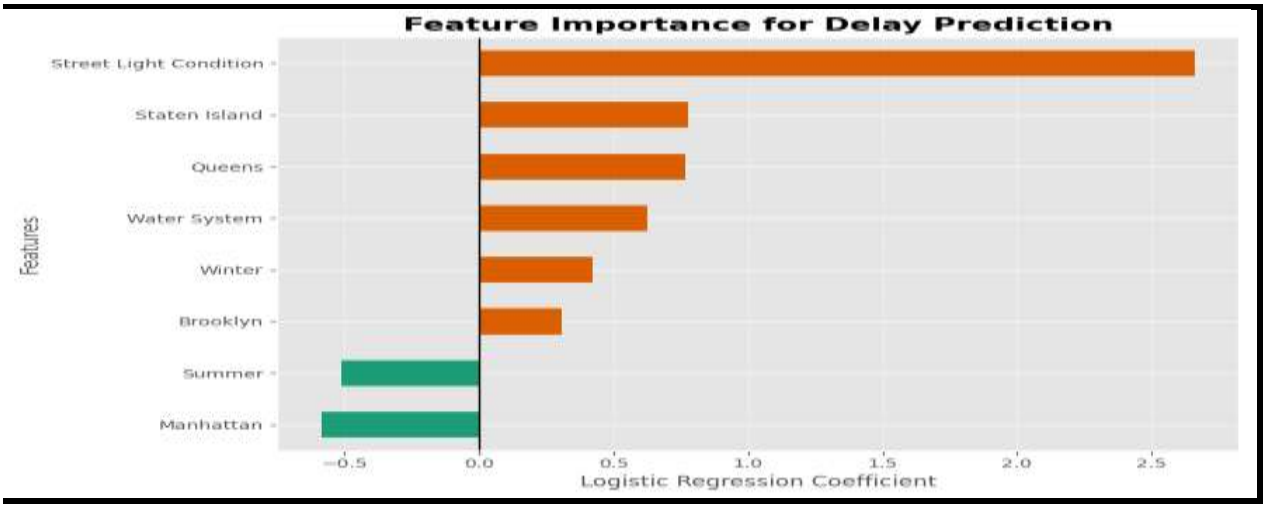

Figure 20: *Logistic Regression Feature Importance*

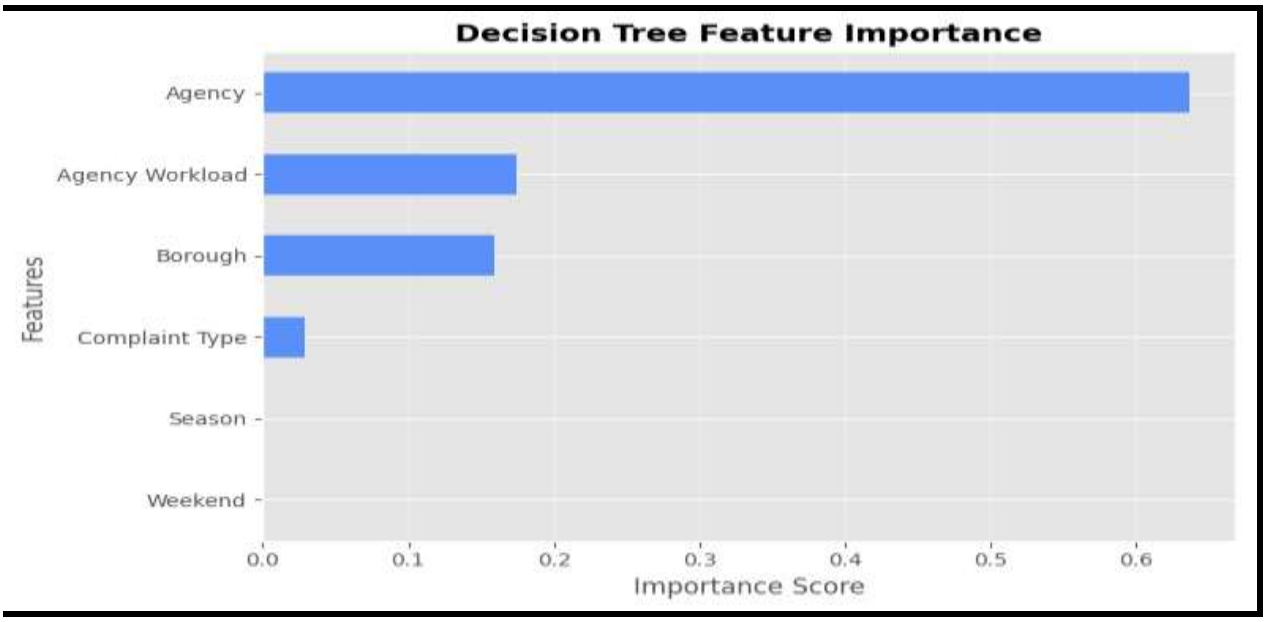

Figure 21*: Decision Tree Feature Importance*

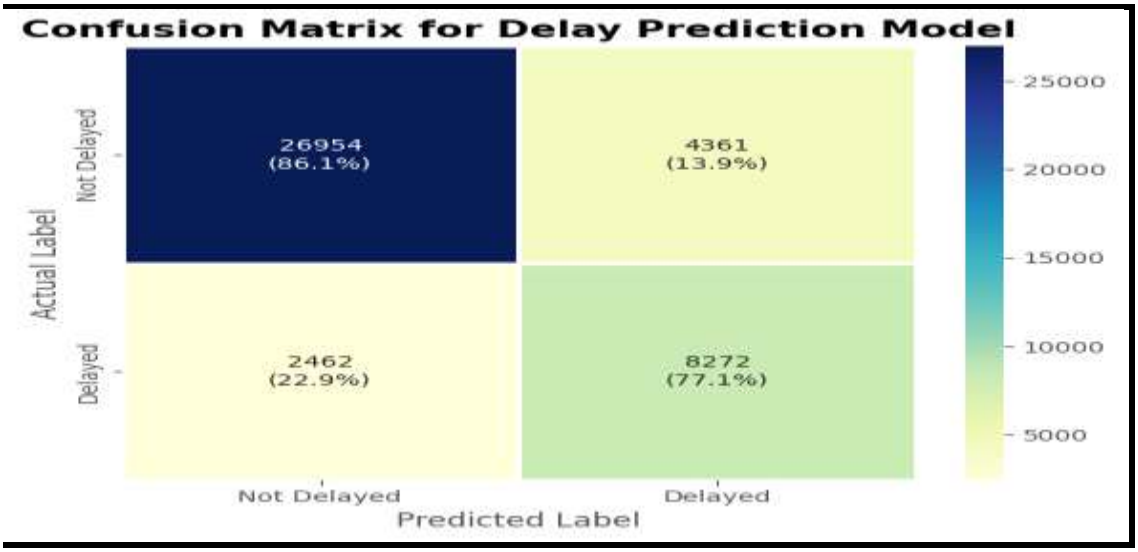

Figure 22: *Confusion Matrix for Delay Prediction model*

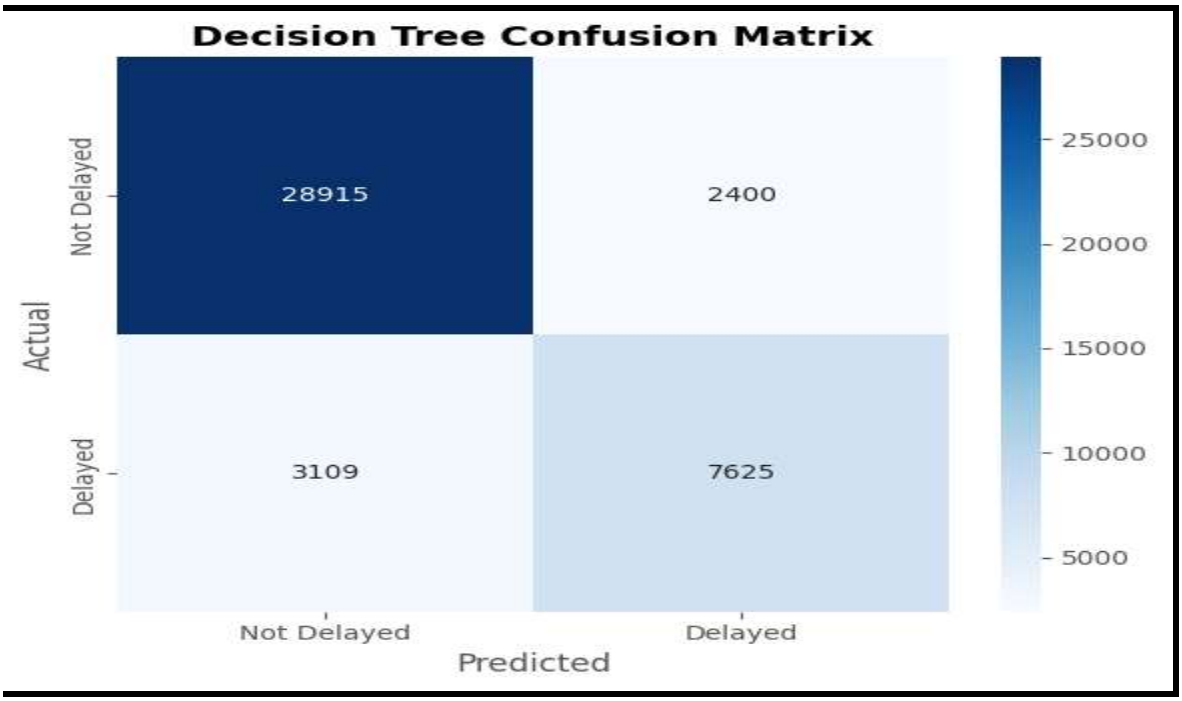

Figure 23: *Decision Tree Confusion Matrix*

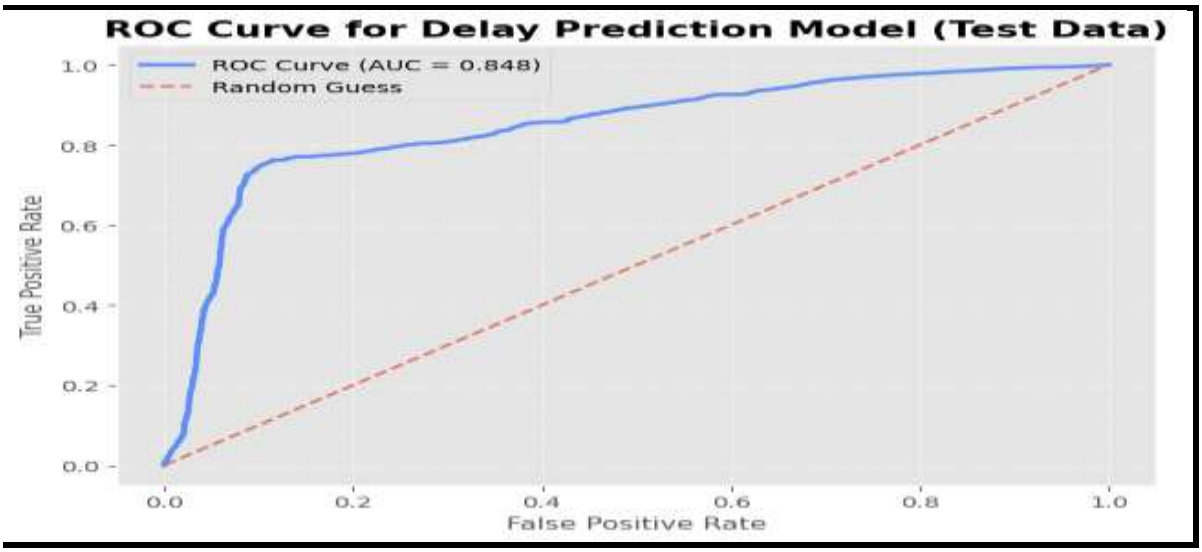

Figure 24: *ROC Curve For Delay Prediction Model*

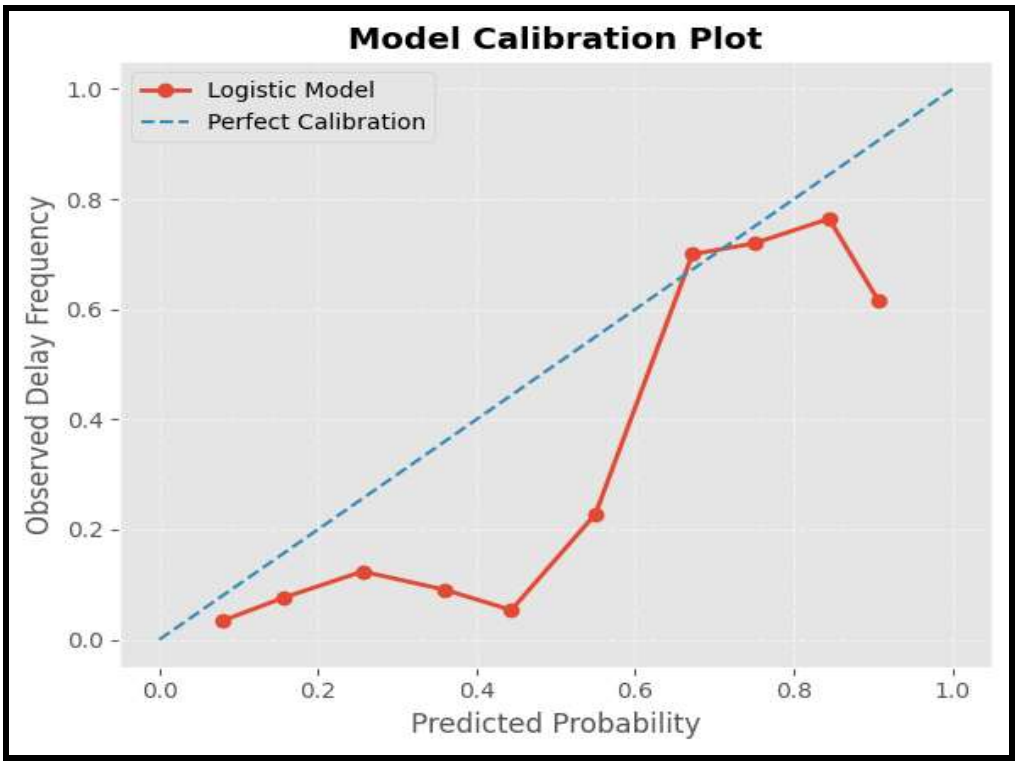

Figure 25: *Model Calibration Plot*

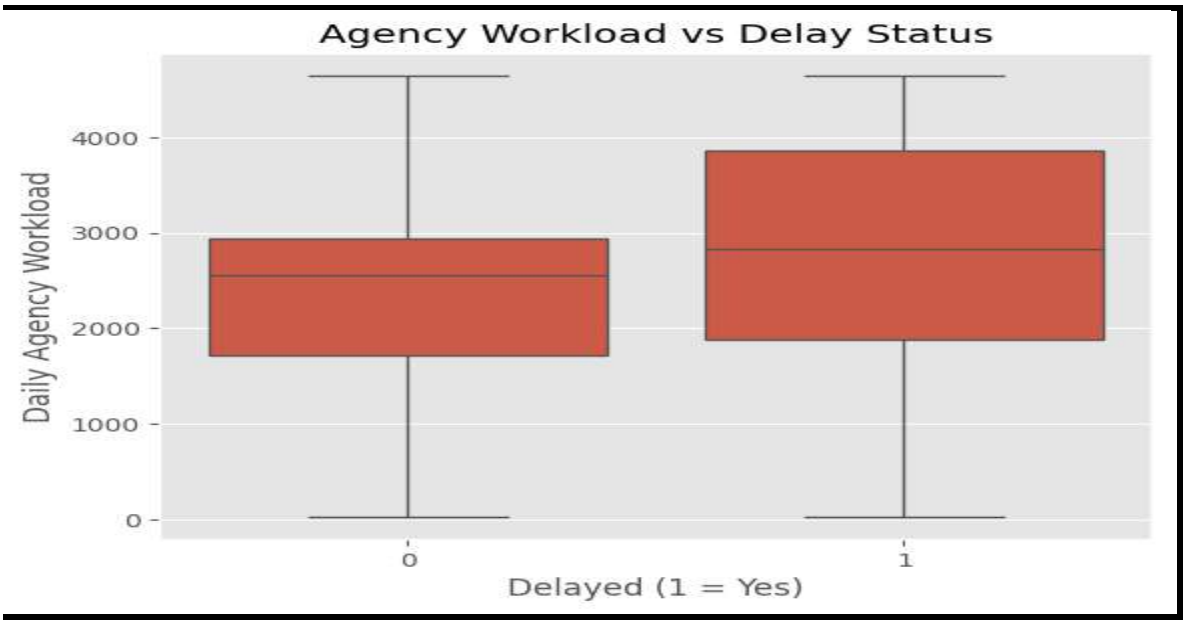

Figure 26: *Agency Workload vs Delay Boxplot*

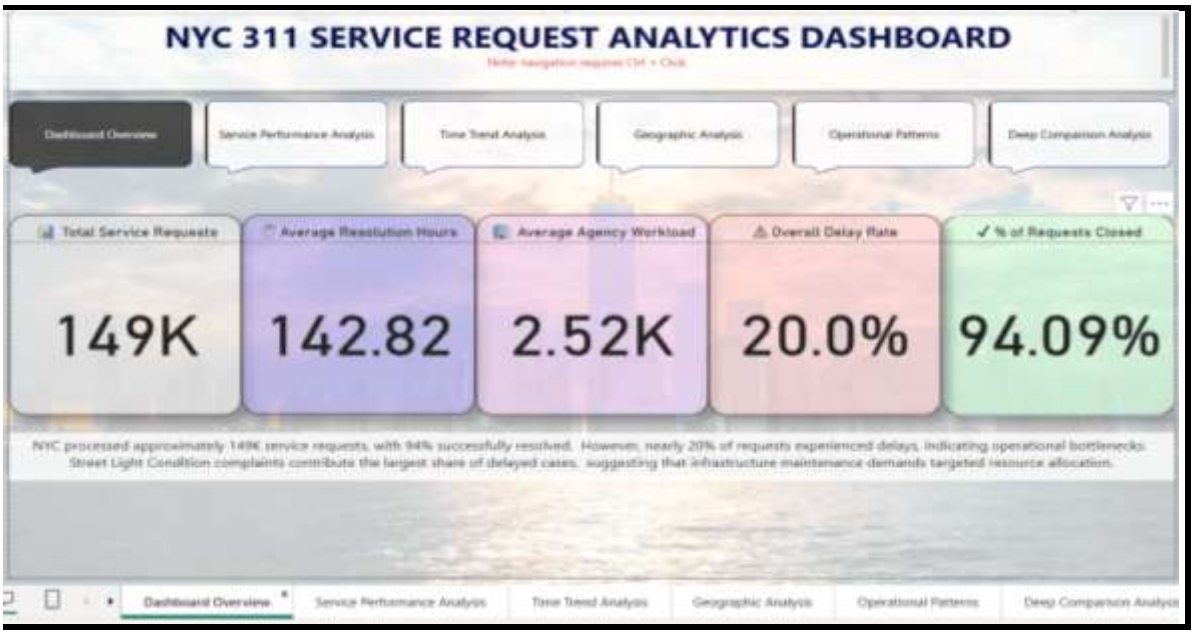

Figure 27: *NYC 311 Service Request Analytics Dashboard Overview*

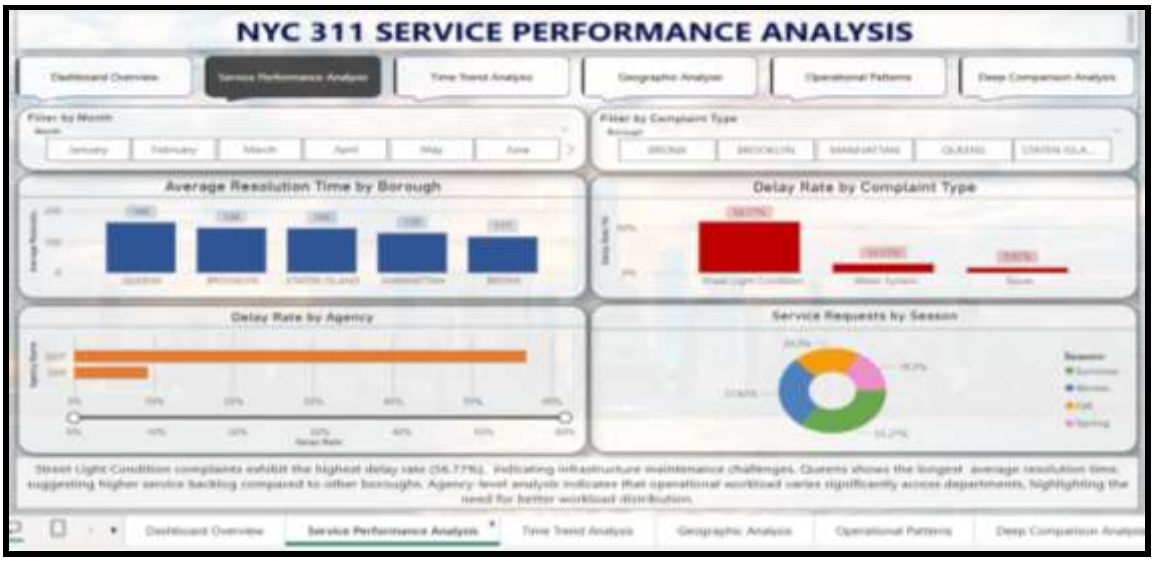

Figure 28: *Service Performance Analysis Dashboard*

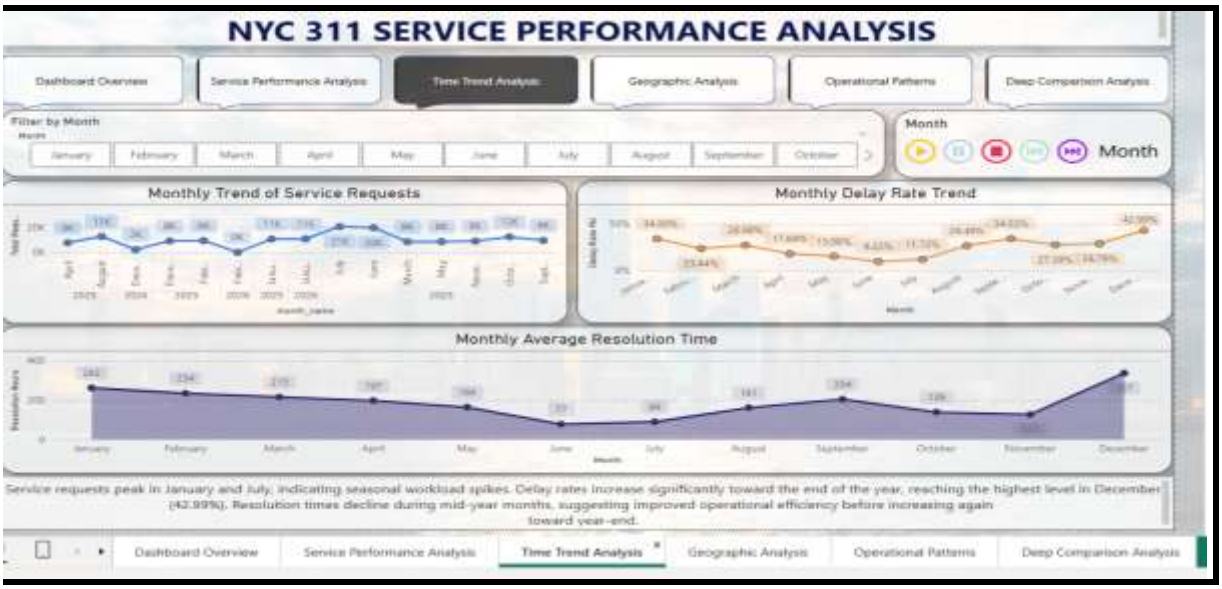

Figure 29: Time Trend Analysis Dashboard

Figure 30: *Geographic Analysis Dashboard*

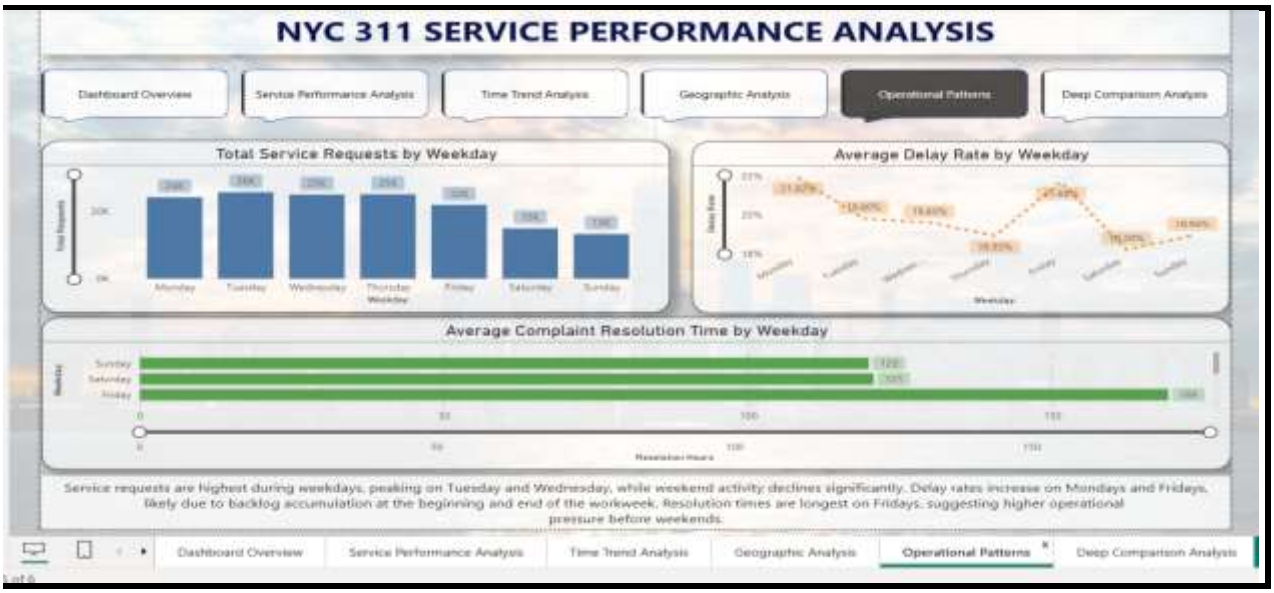

Figure 31: *Operational Patterns Dashboard*

Figure 32: *Deep Comparison Analysis Dashboard*

The Power BI dashboard developed in this study is available through the following Google Drive repository.

Google Drive Link: Power BI Dashboard

The repository includes the .pbix file, which contains the complete interactive dashboard with all data models, visualizations, and analytical components.

In addition, a PDF version of the dashboard is provided for easy viewing and quick reference, allowing the results to be reviewed without requiring Power BI software

5. APPENDIX E: POSTER

The final poster developed for this capstone project is available via the following Google Drive link for review and reference: Capstone Poster

6. APPENDIX F: PRESENTATION

The presentation developed for this capstone project is available via the following Google Drive link for review and reference: Capstone Presentation